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Providentia

An evaluation software package for the
atmospheric modelling community

22/07/2026

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Introduction



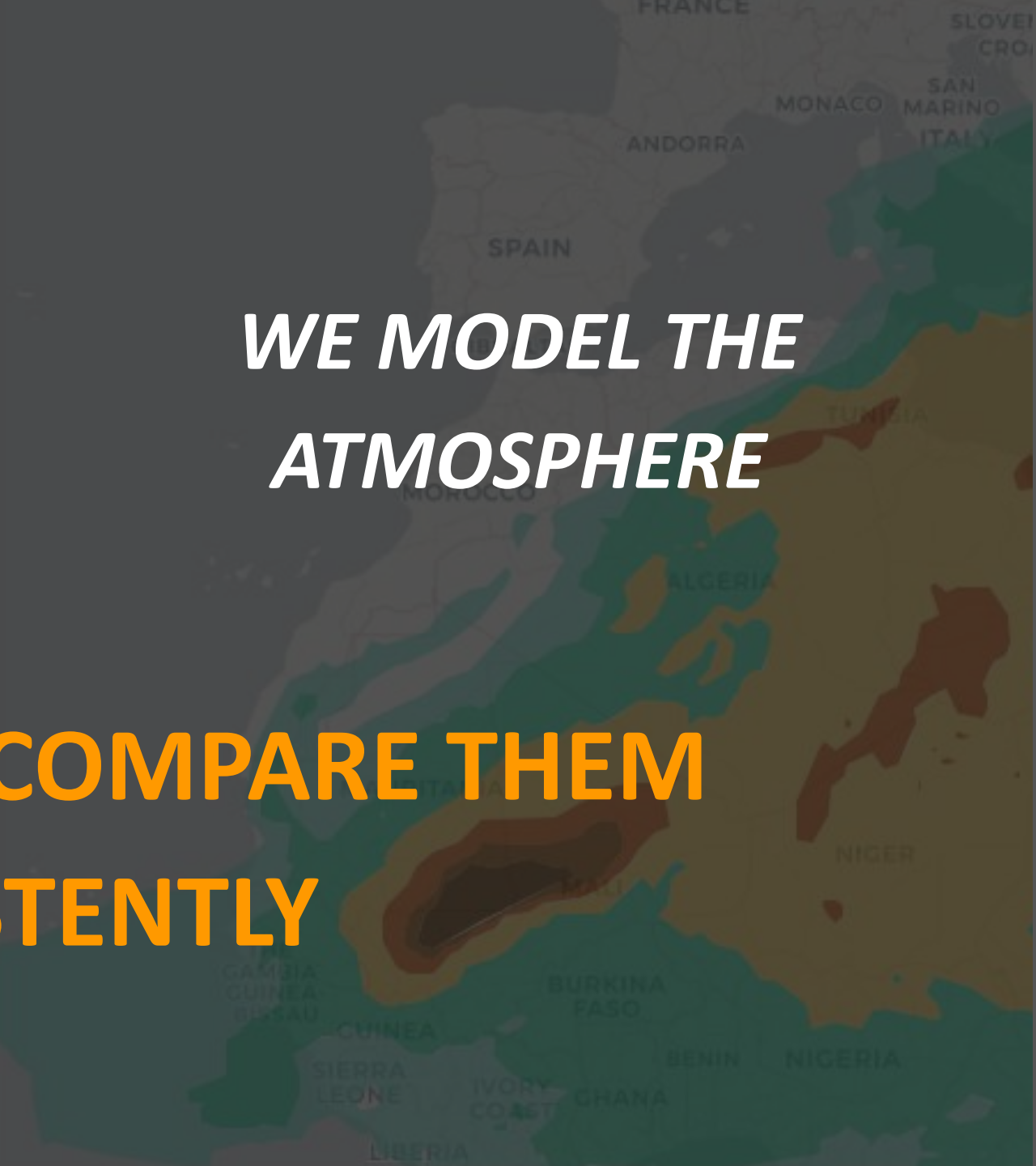
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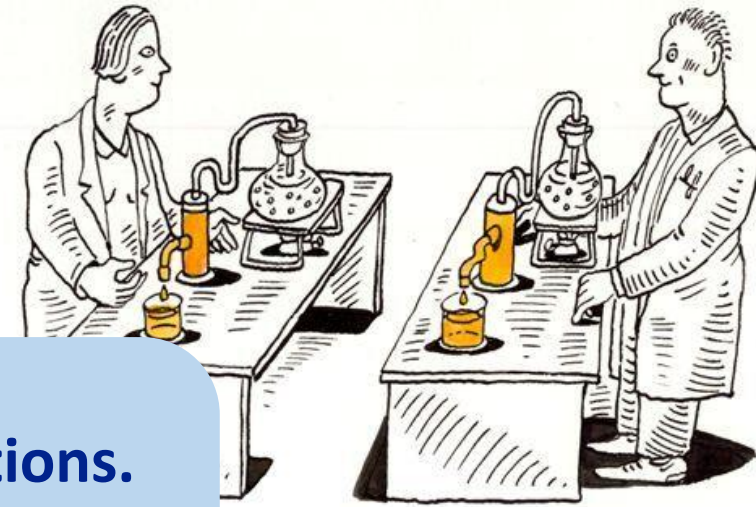
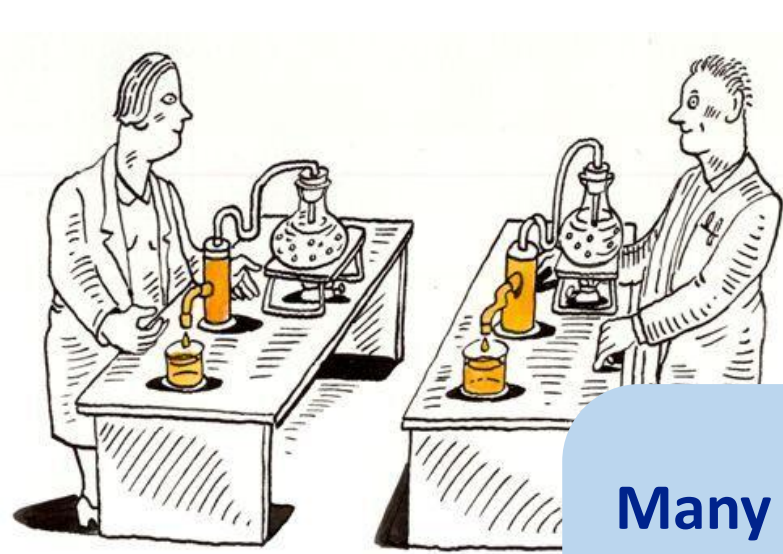


***WE MEASURE THE
ATMOSPHERE***

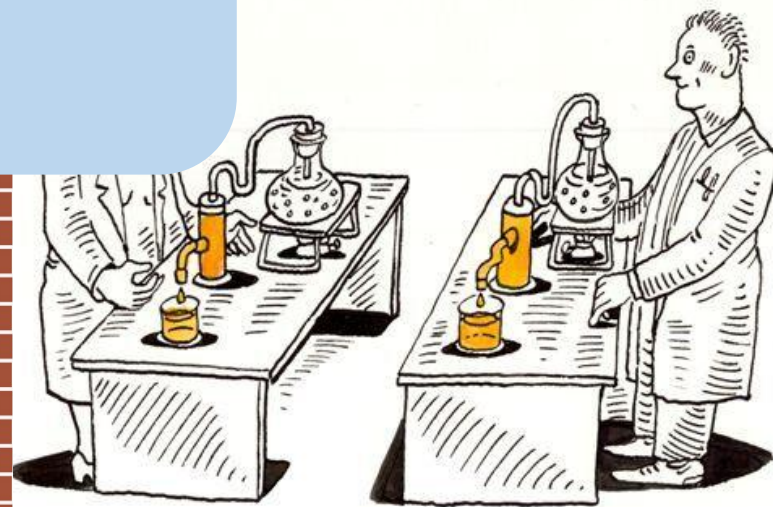
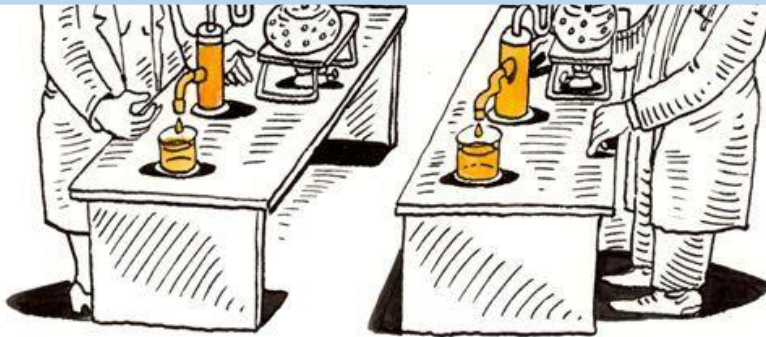
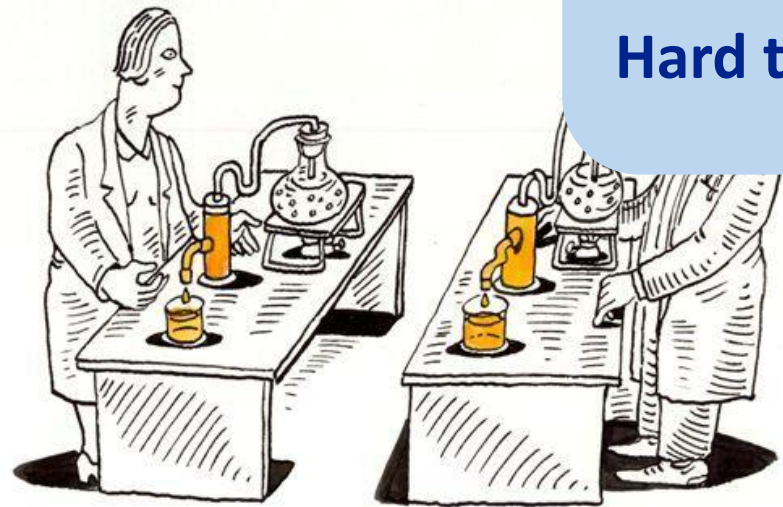
**BUT WE DON'T COMPARE THEM
CONSISTENTLY**

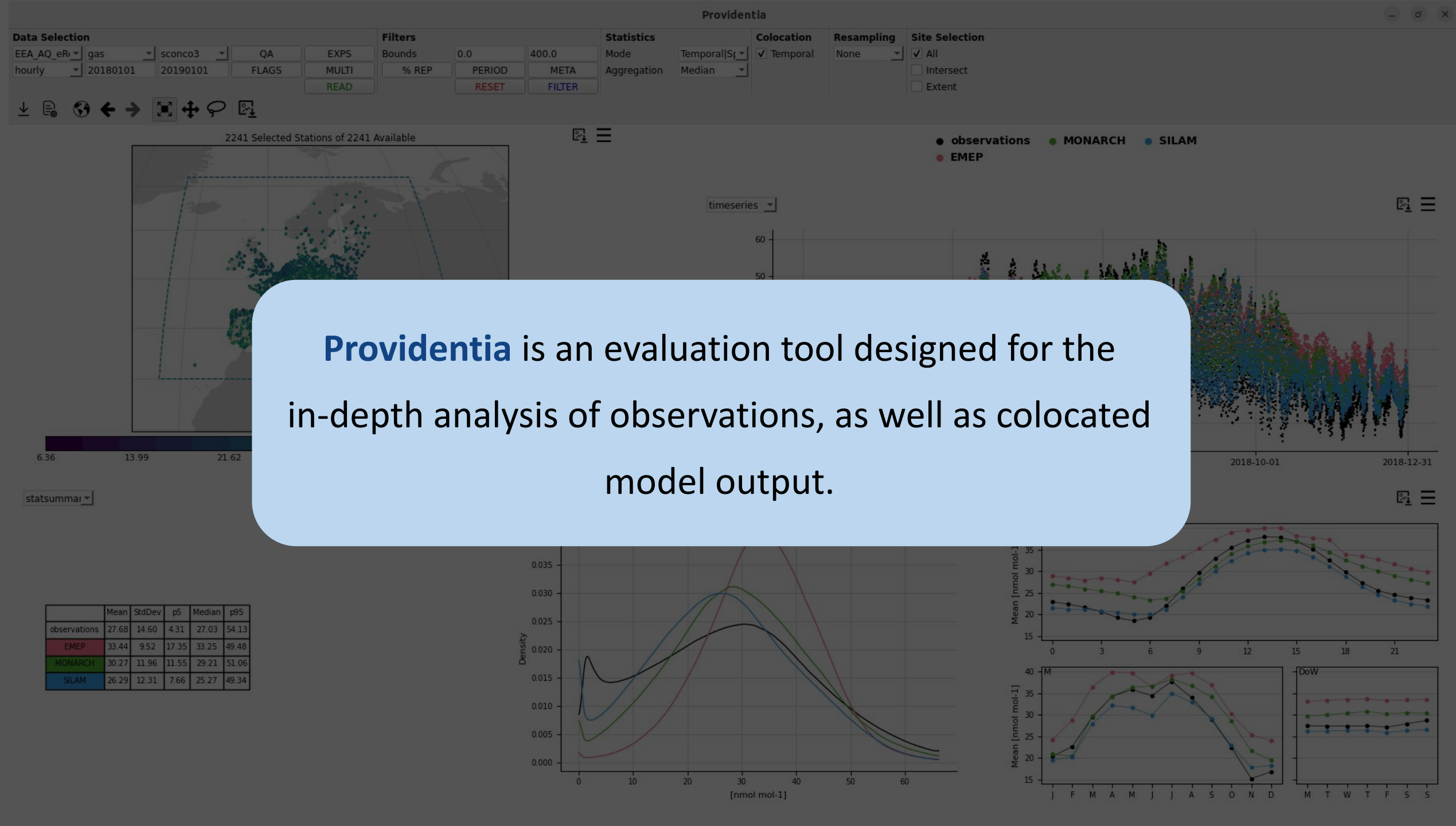


***WE MODEL THE
ATMOSPHERE***



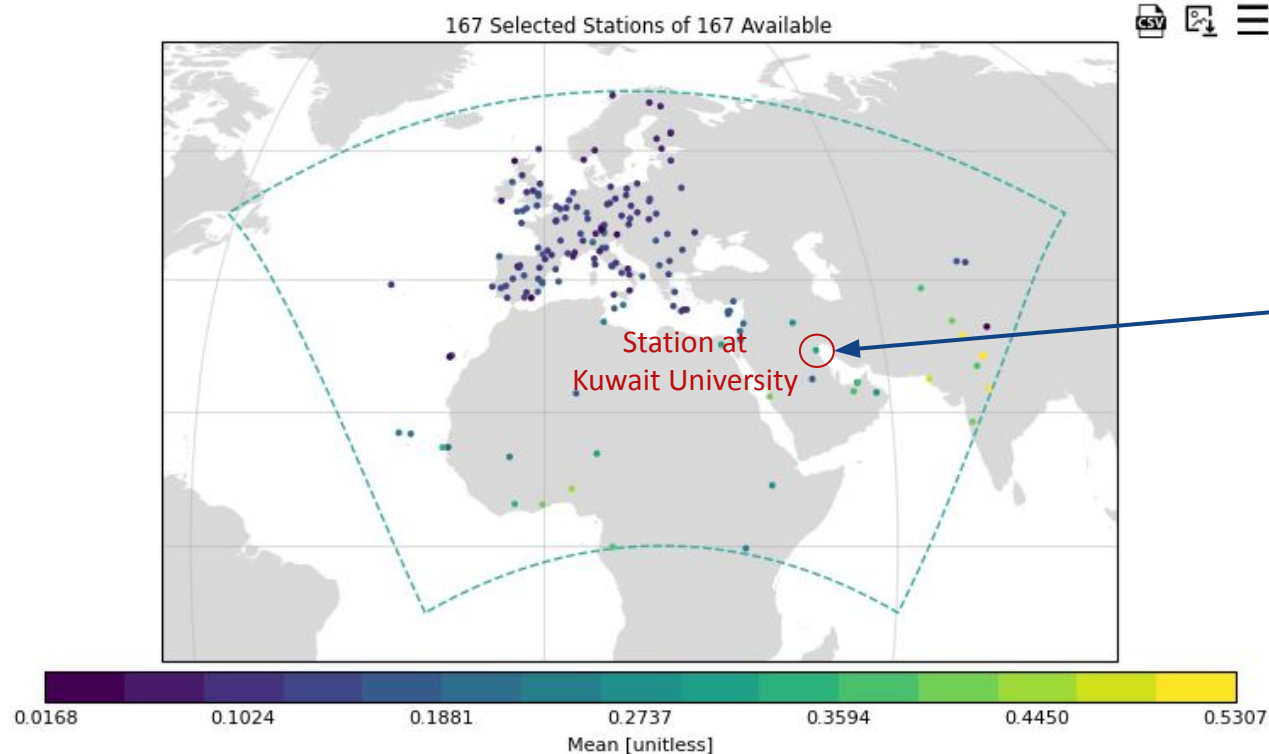
**Many try to combine models and observations.
But each group does it differently.
Hard to reproduce. Hard to compare.**





Why was Providentia created?

Reason 1: Evaluate ground-based station data



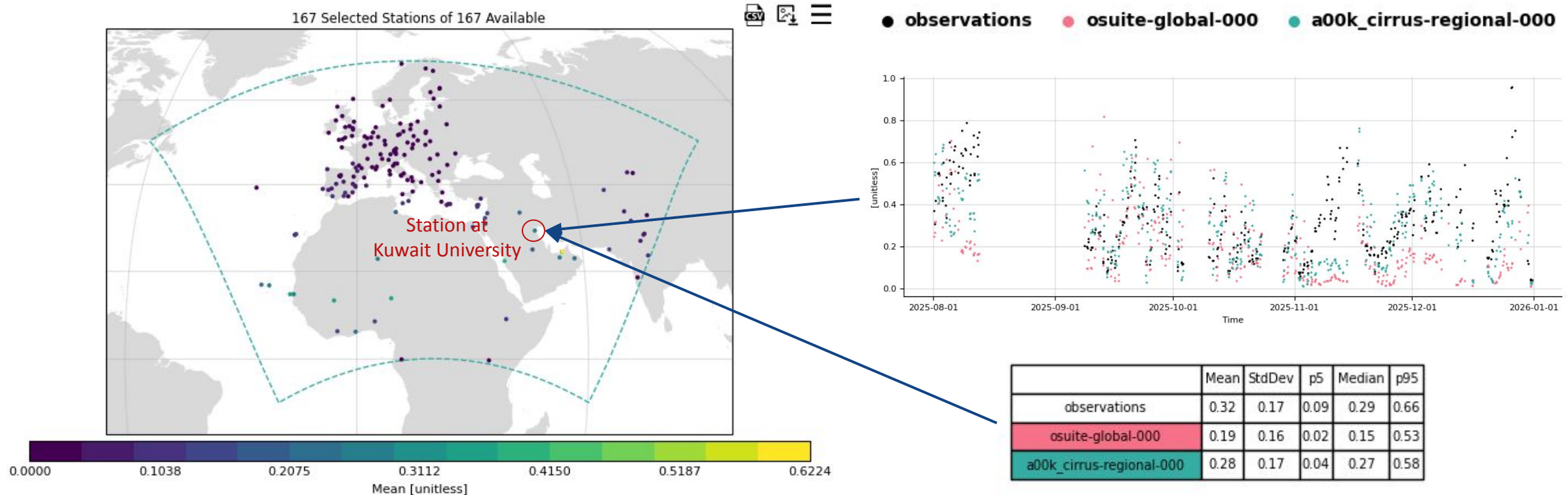
AERONET sun photometer

To measure direct sunlight and sky radiances and multiple wavelengths. These measurements are used to calculate key atmospheric properties, primarily Aerosol Optical Depth (AOD).

Mean of the **observed** surface concentration of aod550aero (July - December 2025)

Why was Providentia created?

Reason 2: Evaluate model data using ground-based station data to **improve forecasts**



Mean of the **modelled** surface concentration of aod550du (July - December 2025)

Structure

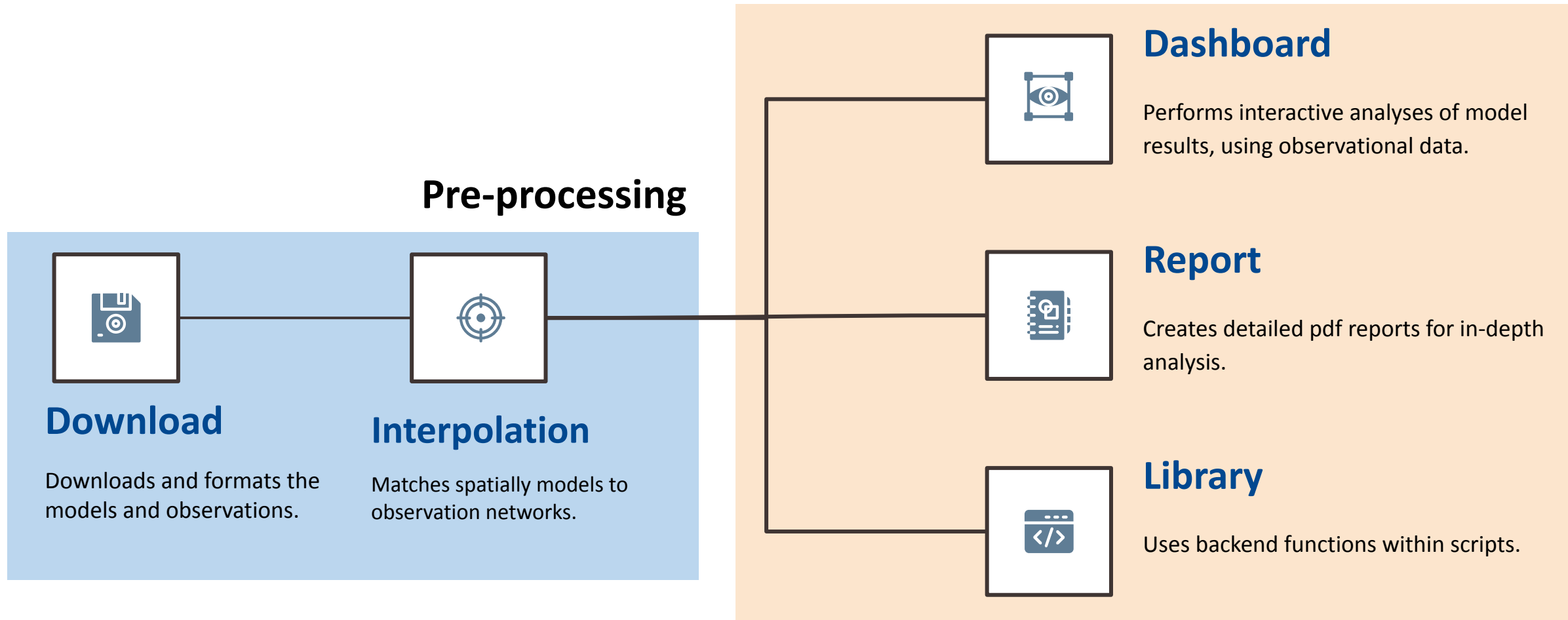


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Structure

Analysis



Dashboard

Data Selection

γ_v3-lev15

3hourly

aod

20250701

od550aero

20260101

QA

FLAGS

MODELS

SPECIES

READ

Filters

Bounds

0.0

1

COVERAGE

PERIOD

RESET

METADATA

FILTER

Statistics

Mode

Temporal

Aggregation

Median

Colocation

Temporal

Resampling

None

Site Selection

All

Intersect

Extent

Download

Print

Fullscreen

Zoom In

Zoom Out

Reset

Help

167 Selected Stations of 167 Available

0.0000 0.1038 0.2075 0.3112 0.4150 0.5187 0.6224

Mean [unitless]

statsumme

	Mean	StdDev	p5	Median	p95
observations	0.13	0.09	0.04	0.09	0.32
osuite-global-000	0.01	0.03	2.79e-04	0.00	0.06
a00k_cirrus-regional-000	0.03	0.05	4.24e-08	0.00	0.14

distributor

● observations ● osuite-global-000 ● a00k_cirrus-regional-000

timeseries

[unitless]

Time

periodic

Mean [unitless]

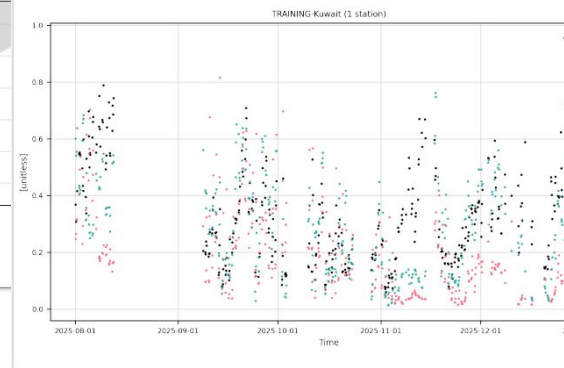
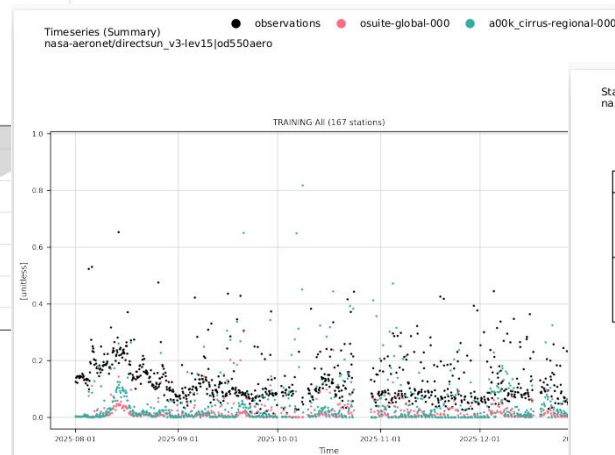
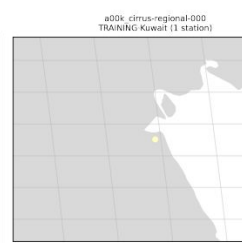
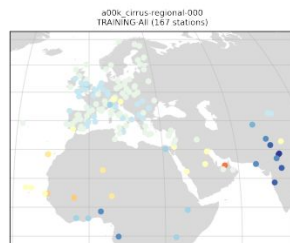
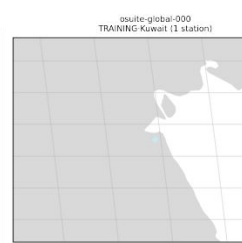
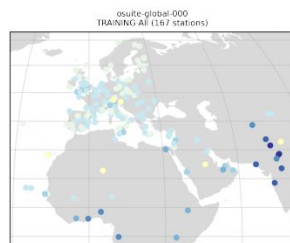
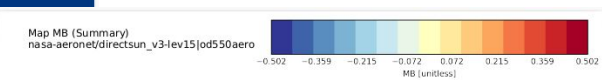
A S O N D

M T W T F S S

Report

EVALUATION REPORT

Network: nasa-aeronet/directsun_v3-lev15
Species: AOD_550nm
Temporal resolution: 3hourly
Date range: 01/07/2025 - 01/01/2026
Models: osuite-global-000, a00k_cirrus-regional-000
Subsections: TRAINING All, TRAINING-Kuwait



StatSummary (Summary)
nasa-aeronet/directsun_v3-lev15(od550aero)

		Mean	StdDev	p5	Median	p95
All	observations	0.13	0.09	0.04	0.09	0.32
	osuite-global-000	0.01	0.03	2.79e-04	0.00	0.06
	a00k_cirrus-regional-000	0.03	0.05	4.24e-08	0.00	0.14
Kuwait	observations	0.32	0.17	0.09	0.29	0.66
	osuite-global-000	0.19	0.16	0.02	0.15	0.53
	a00k_cirrus-regional-000	0.28	0.17	0.04	0.27	0.58

Library

```
# import Providentia
import providentia as prv
```

```
# magic to make plots appear inline
%matplotlib inline
```

Initialise object

```
provi = prv.Providentia('trainings/20260722_Kuwait_Training_Dust_Modelling/kuwait.conf')
```

Warning: Using the first defined subsection (TRAINING-All).

```
provi.print_config()
```

```
[TRAINING]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000, a00k_cirrus-regional-000
upper_bound = 1
```

```
[[All]]
```

```
[[Kuwait]]
station_reference = Kuwait_University_2
```

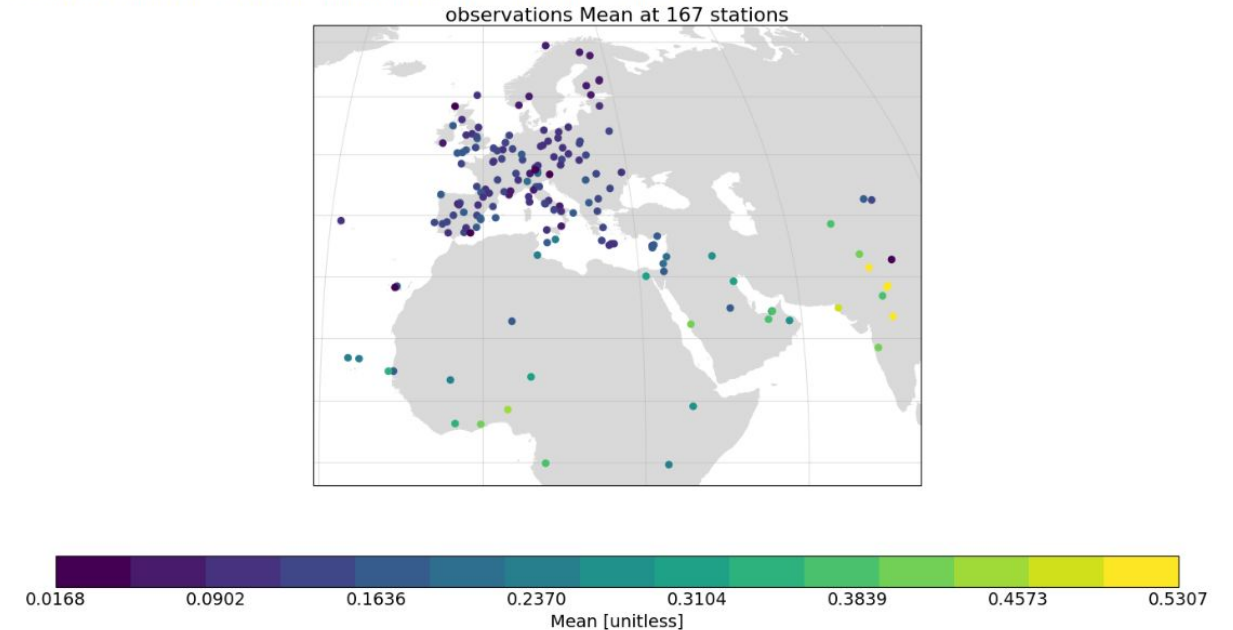
Load data

```
provi.load()
```

Make plots

```
# make a map plot (mean of obs)
provi.plot('map-Mean', labela='observations')
```

Warning: Width and/or height have not been passed. The default values will be set.



Complete tutorial: https://github.com/BSC-ES/providentia/blob/master/trainings/20260722_Kuwait_Training_Dust_Modelling.ipynb

Set up



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Prerequisites

Providentia is stable on macOS and Linux, where you need to install [Git](#) and [Conda](#).

The software has not been designed to work on Windows. If you are a Windows user you must first choose one of these options:

- Use Windows Subsystem for Linux (WSL), only for Windows Pro users (**Recommended**). Instructions [here](#).
- Use a virtual machine (e.g. Oracle, VirtualBox). Instructions [here](#).
- Use Git Bash. Instructions [here](#).

For more details, see the official documentation:

<https://providentia.readthedocs.io/en/latest/Getting-started.html#prerequisites>

Clone Providentia

Enter the project's GitHub page (<https://github.com/bsc-es/providentia>) and clone Providentia in your working directory:

```
git clone https://github.com/bsc-es/providentia
```

The latest releases are saved in the production branch, but we suggest users to stay in the **master** branch. In order to stay up-to-date just run this line from time to time:

```
git pull
```


Launching Providentia

After cloning the repository, open a terminal and move to the Providentia directory.

```
cd providentia
```

Then start the dashboard by running:

```
./bin/providentia
```

The first time the software runs in a local machine it will create a conda environment called **providentia-env_v[version]** with all the modules needed.

Main launch options

	Main purpose	Argument
Dashboard	On-the fly analysis	
Report	Detailed analysis in PDFs format	--report, --reports
Notebook	Open a Jupyter notebook	--notebook, --nb
Interpolation	Interpolate models to observational networks	--interp, --interpolate, --interpolation
Download	Download data to local machine	--download, --dl
Clean	Clean logs and new files	--clean
Configuration file	Path or name (if in configurations/ directory)	--conf, --config, --configuration
Section	Specify configuration file section	--section
Logfile	Redirects output to a file.	--logfile

For more details, see the official documentation:

<https://providentia.readthedocs.io/en/latest/Command-line-configuration.html>

Data

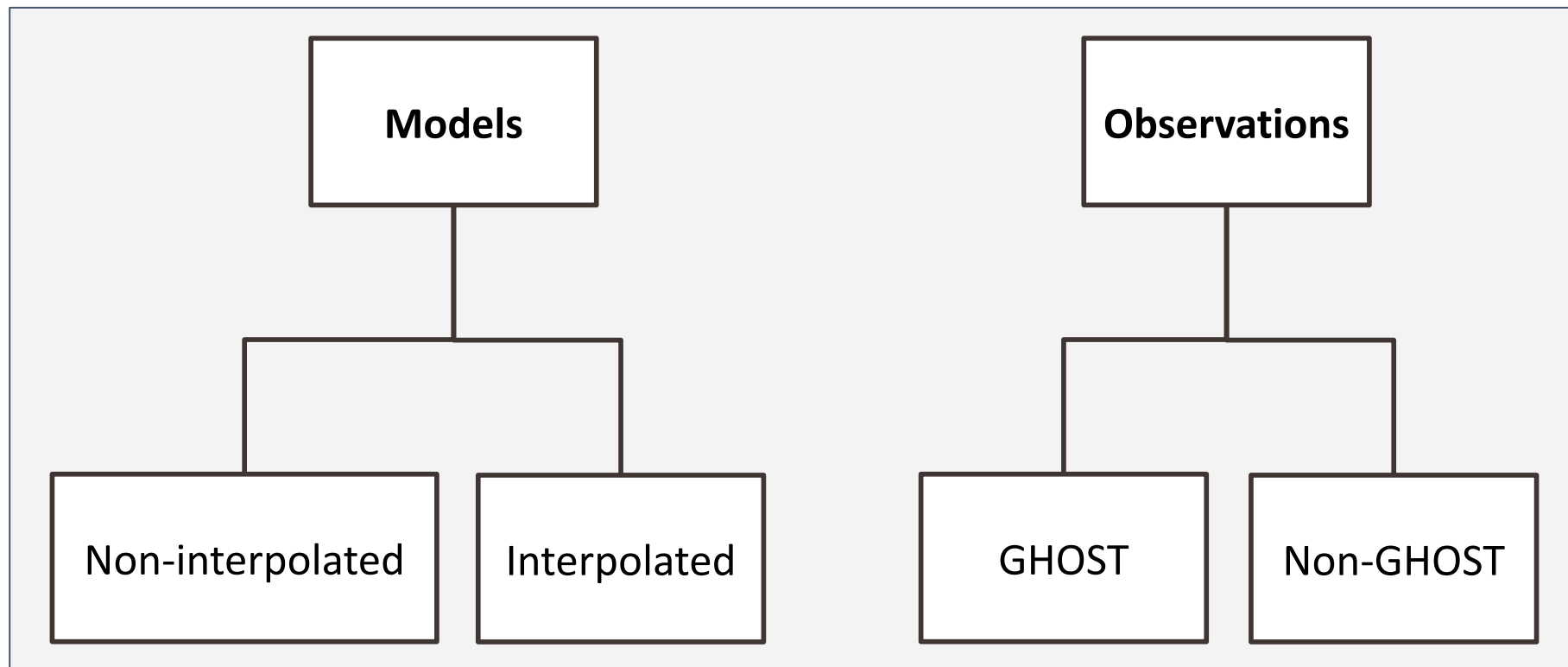


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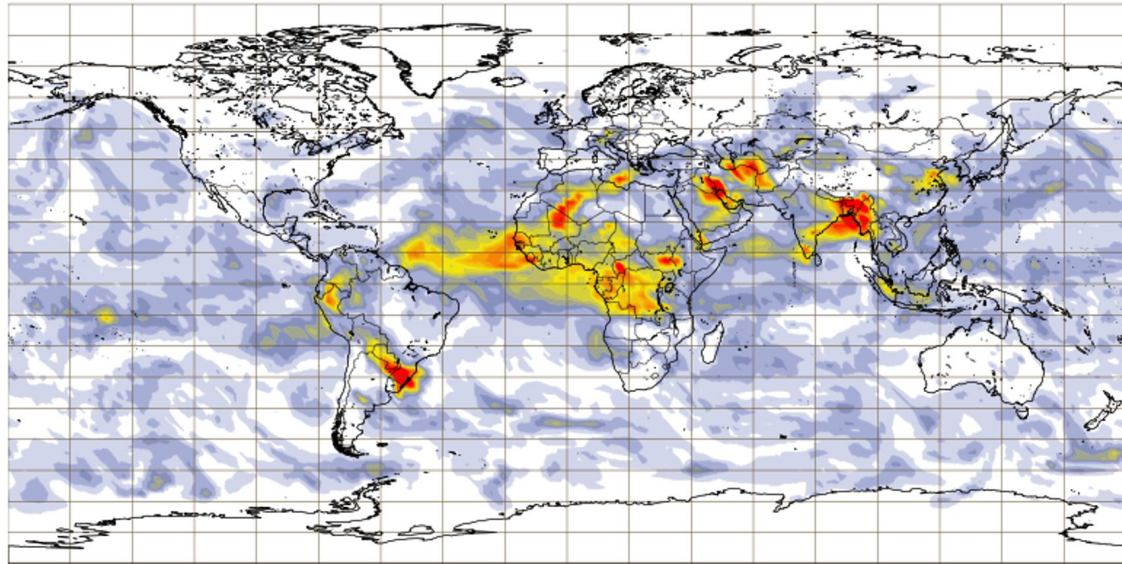
Data classification

To help you understand the data used in Providentia, we classify it into the following groups and subgroups:



Non-interpolated model

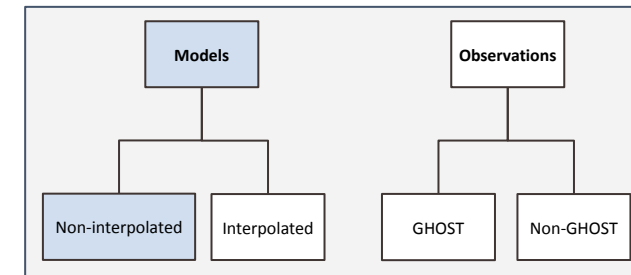
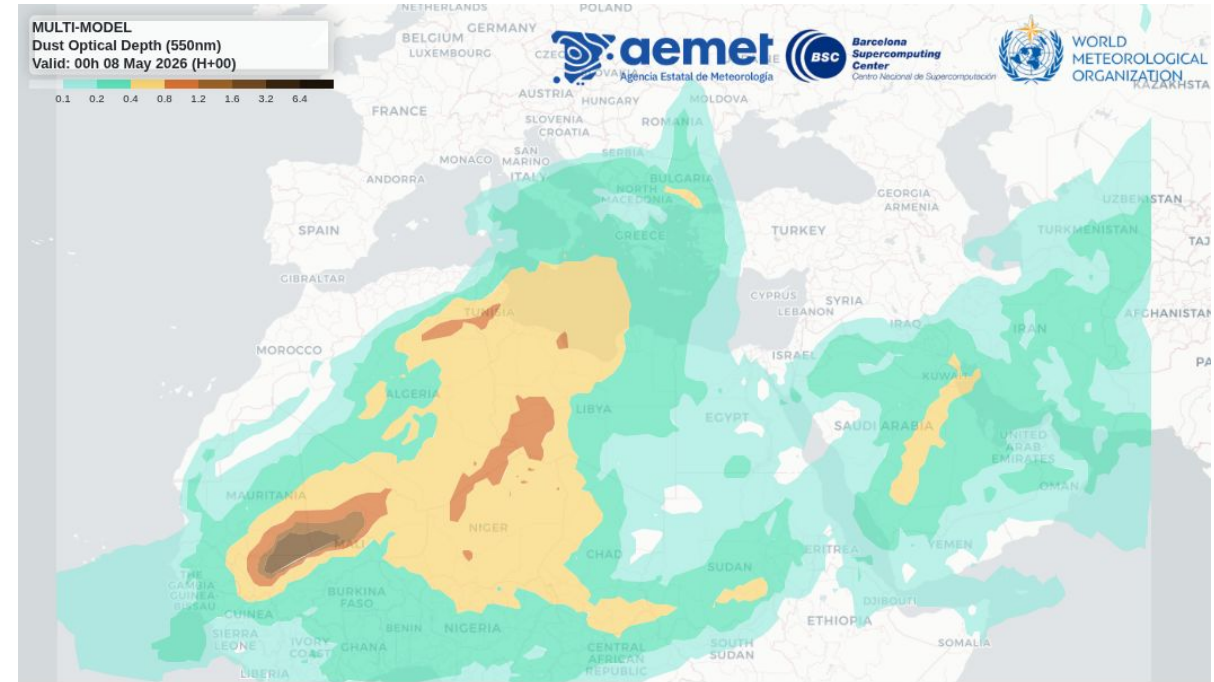
CAMS AOD



Aerosol optical depth at 550 nm (provided by CAMS, the Copernicus Atmosphere Monitoring Service)

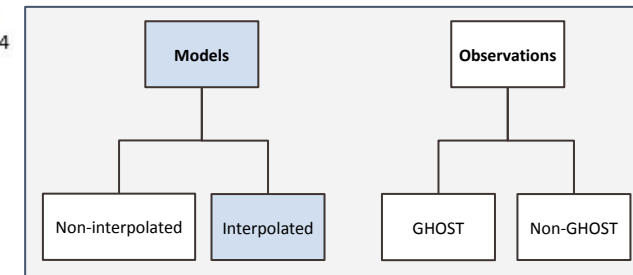
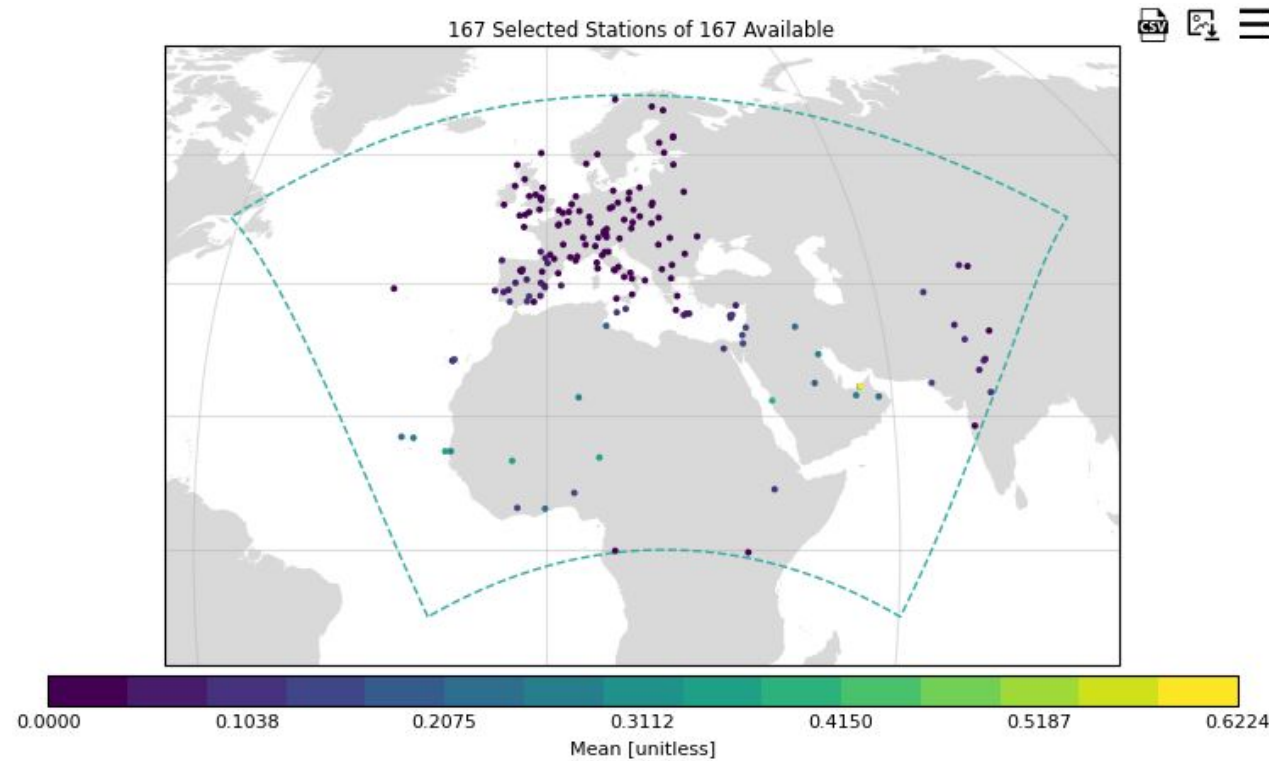
CAMS aerosol forecasts

BDRC DOD



Interpolated model

Interpolation in Providentia refers to the mathematical process of estimating the model variable at the exact location of an observational station. *More to come in the following slides.*



GHOST:

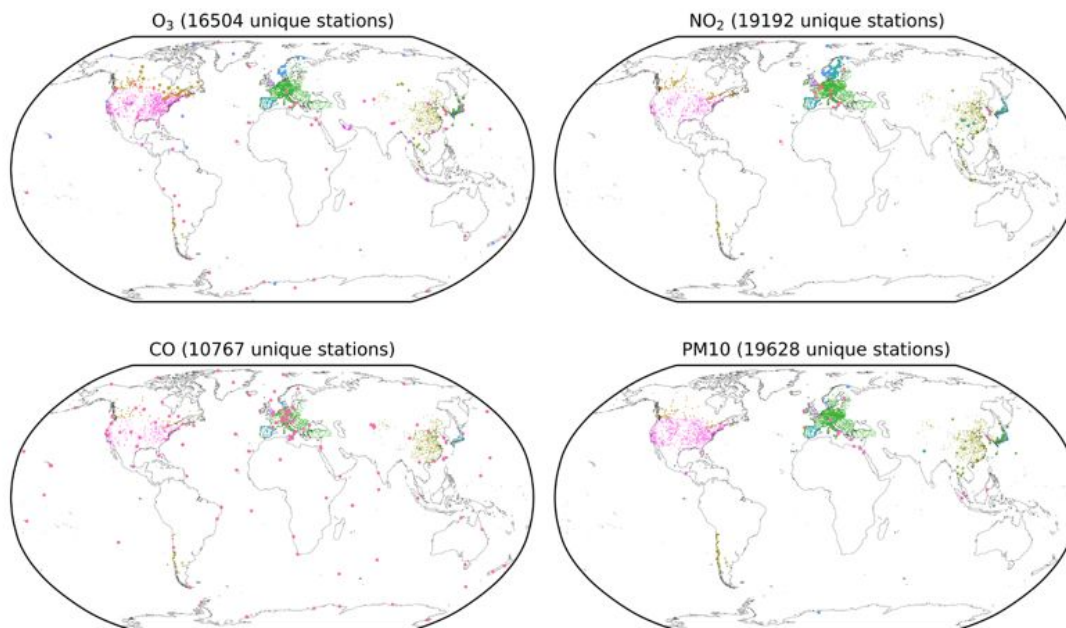
Globally harmonised observations in space and time

N° Processed
Measurements:
~7.3 billion

N° Components:
227

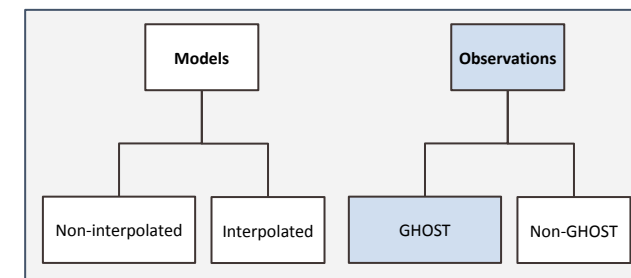
N° Networks:
38

Temporal range:
1970-2023



- | | |
|------------------------|------------------------|
| ● ACTRIS | ● AERONET v3 Level 1.5 |
| ● BJMEMC | ● CAMP |
| ● Chile SINCA | ● CNEMC |
| ● EEA AirBase | ● EEA AQ e-Reporting |
| ● EUSAAR | ● HELCOM |
| ● Independent (EBAS) | ● Japan NIES |
| ● NADP AMNet | ● NADP AMoN |
| ● NOAA-GGGRN | ● OECD |
| ● US EPA AirNow DOS | ● US EPA AQS |
| ● WMO GAW WDCGG | ● WMO GAW WDCRG |
| ● AERONET v3 Level 2.0 | ● AMAP |
| ● Canada NAPS | ● CAPMoN |
| ● COLOSSAL | ● EANET |
| ● EMEP | ● EUCAARI |
| ● HTAP | ● IMPACTS |
| ● Mexico CDMX | ● MITECO |
| ● NILU | ● NOAA-ESRL |
| ● UK AIR | ● UK DECC |
| ● US EPA CASTNET | ● WMO GAW WDCA |

Bowdalo, D., Basart, S., Guevara, M., Jorba, O., Pérez García-Pando, C., Jaimes Palomera, M., Rivera Hernandez, O., Puchalski, M., Gay, D., Klausen, J., Moreno, S., Netcheva, S., and Tarasova, O.: GHOST: a globally harmonised dataset of surface atmospheric composition measurements, *Earth Syst. Sci. Data*, 16, 4417–4495, <https://doi.org/10.5194/essd-16-4417-2024>, 2024.



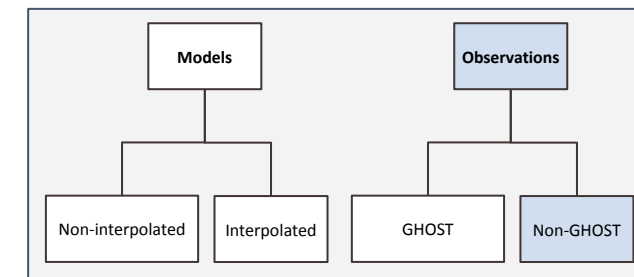
non-GHOST

Providentia can read data non-harmonised observational data, named as non-GHOST, the limitation is the data will be have less filtering features, specifically:

- No GHOST QA flags
- No network flags
- No native representativity filters
- No period filters
- Limited metadata

On the dashboard this is reflected via greyed out buttons:

Data Selection				Filters		
r_v3-lev15	aod	od550aero	QA	MODELS	0.0	1
3hourly	20250701	20260101	FLAGS	SPECIES	COVERAGE	PERIOD
				READ	RESET	FILTER



GHOST vs. non-GHOST datasets

	GHOST	non-GHOST
Data availability	Public in Zenodo until 2023	Most recent
Accessibility	Zenodo	Non-formatted directly from data portals (e.g. AERONET from https://aeronet.gsfc.nasa.gov/)
How to download	Providentia download mode	Dependent on provider (e.g. AERONET from user interface or client API)
Metadata	✓	✗
Quality assurance	✓	✗

Setting data paths

Providentia knows where data lives via one key file: **settings/data_paths.yaml**

There are multiple headings in the file which set **observational** and **model** data paths on different machines, e.g. **local** and BSC machines (**mn5**, **nord4**, etc.).

Under each heading, the following paths are set:

- **ghost_root** : Observational GHOST data
- **nonghost_root**: Observational non-GHOST data
- **mod_root**: Interpolated model data
- **mod_to_interp_root**: Non-interpolated model data

```
local:  
  ghost_root: ~/data/providentia/obs/ghost  
  mod_root: ~/data/providentia/mod  
  mod_to_interp_root: ~/data/providentia/mod_to_interp  
  nonghost_root: ~/data/providentia/obs/nonghost
```

If you are using Linux inside Windows (through WSL, VirtualBox or Git Bash) you will need to modify the paths to point to your Windows paths.

Data directory format

Depending on the type of data, the directory structure changes:

- **Observational Data**

- **GHOST**

- {ghost_root}/{network}/{ghost_version}/{resolution}/{species}/{species_YYYYMM.nc}

- **Non-GHOST**

- {nonghost_root}/{institution}/{network}/{resolution}/{species}/{species_YYYYMM.nc}

- **Model Data**

- **Non-Interpolated**

- {mod_to_interp_root}/{model_id}/{domain}/{resolution}/{species}/{species_YYYYMM[DD].nc}

- **Interpolated**

- {mod_root}/{ghost_version}/{model_id}-{domain}-{ensemble}/{resolution}/{species}/{network}/{species_YYYYMM.nc}

Configuration files



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Create a configuration file

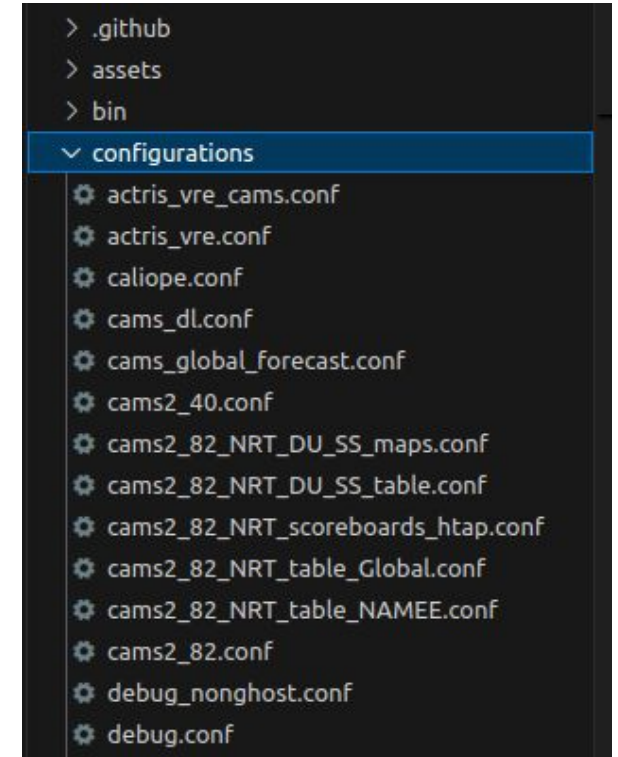
Configuration files are files where the user **defines the parameters** used to specify which data to:

- Download (download mode)
- Interpolate (interpolation mode)
- Read and plot (dashboard, report and library modes)

By default, Providentia reads these files from the **configurations/** directory.

For more details, see the official documentation:

<https://providentia.readthedocs.io/en/latest/Configuration-files.html>



Using a configuration file

To specify which configuration file to use, you can specify it from the command line using the **--config** argument.

Configuration file inside the configurations/ folder

If the file is located inside the **configurations/** folder, only the filename is required:

```
./bin/providentia --config=example.conf
```

Configuration file in a different location:

If the file is stored in a different location, you can provide the full path:

```
/bin/providentia --config=/path/to/file/example.conf
```


Main fields in a configuration file

The following fields are commonly defined in configuration files across all the different modes:

	Description
network	Observational data. (e.g. nasa-aeronet/directsun_v3-lev15)
models, model	Model IDs. Domain and ensemble information can also be included. (e.g. a00k_cirrus-regional-000)
species	Atmospheric components to analyse in BSC nomenclature. (e.g. od550aero)
resolution	Temporal resolution of the observations (e.g. 3hourly)
start_date	Start date. In YYYYMMDD format or YYYYMM for interpolation. (e.g. 20250701)
end_date	End date. In YYYYMMDD format or YYYYMM for interpolation. (e.g. 20260101)

Species

- Species refer to atmospheric components such as gases and aerosols.
- Providentia uses the nomenclature defined in [GHOST standards](#).
- AERONET parameter names can be read in Providentia directly.

Description	AERONET parameter name
Total aerosol optical depth at 550nm	od550aero
Total coarse aerosol optical depth at 500nm	od500aerocoarse
Angstrom exponent at 440-870 nm	ae440-870aero

Model, ensemble and domain

In Providentia, models are defined using three key components:

- **Model:** Model ID (or experiment ID in MONARCH terminology).
Examples: a00c_cirrus-regional-000.
- **Domain:** Geographical region where the model is run.
Examples: global, eu, regional, cat.
- **Ensemble:** Simulation of the same model with different initial conditions or configurations to estimate uncertainty.
Examples: 000, 001, 002.

You can specify **model, domain, and ensemble** either:

- All in one line.
- Separately in different fields.
- A combination of both.

For more details, check the models section in the official documentation: [Models configuration](#)

```
[TRAINING]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000
```

```
[TRAINING]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite
domain = global
ensemble = 000
```

Structure of a configuration file

Configuration files are composed of:

- Sections (mandatory) act as the parent container for all parameters.
- Subsections (optional), contain metadata to filter data.

When reading and filtering data, subsections inherit all values defined in their parent section.

Section [name]

```
[TRAINING]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000, a00k_cirrus-regional-000
```

Subsections [[name]]

```
[[All]]
```

```
[[Kuwait]]
```

```
station_reference = Kuwait_University_2
```

Additional notes:

- Configuration fields are not order-sensitive.
- When defining a section-subsection pair in the command line, use:

--section=section·subsection

Multiple sections in a configuration file

Section1 ([name])

```
[TRAINING-All]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000, a00k_cirrus-regional-000
```

Section1·Subsection1 ([[name]])

```
[[All]]
```

Section2 ([name])

```
[TRAINING-Kuwait]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000, a00k_cirrus-regional-000
```

Section2·Subsection1 ([[name]])

```
[[Kuwait]]
station_reference = Kuwait_University_2
```



BREAK
5 min



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Download



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Download data to your local machine

Providentia provides direct access to observational and model data that is publicly available via a dedicated download mode.

Observational data

- ACTRIS (via NILU THREDDS)
- GHOST (Zenodo)

Model data

- ERA5 reanalysis
- CAMS regional forecast, analysis, and reanalysis
- CAMS global forecast, and reanalysis

For more details, see the official documentation:

<https://providentia.readthedocs.io/en/latest/Download-mode.html>

ACTRIS

The Aerosol, Clouds and Trace
Gases Research Infrastructure

In-situ atmospheric
measurements



GHOST

Globally Harmonised
Observations in Space and Time

N° Processed
Measurements:

~7.3 billion

N° Components:

227

N° Networks:

38

Temporal range:

1970-2023

 Facilities 

 ACTRIS National Facility - In Progress 

ERA5 reanalysis

Weather and climate variables
Global dataset



Climate
Change Service

climate.copernicus.eu

CAMS forecasts and reanalyses

Atmospheric composition variables
Global + European dataset



JÜLICH
Forschungszentrum



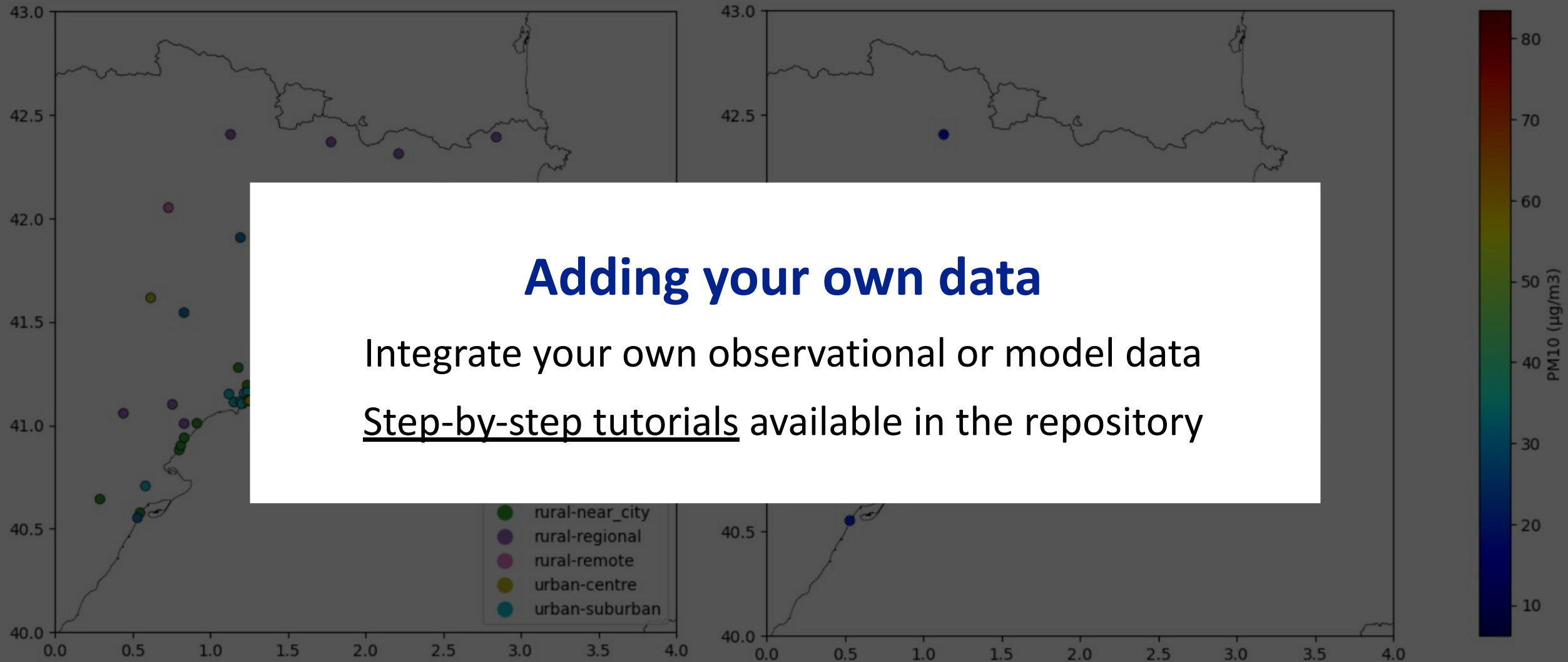
Atmosphere
Monitoring Service

atmosphere.copernicus.eu

Adding your own data

Integrate your own observational or model data

Step-by-step tutorials available in the repository



Interpolation



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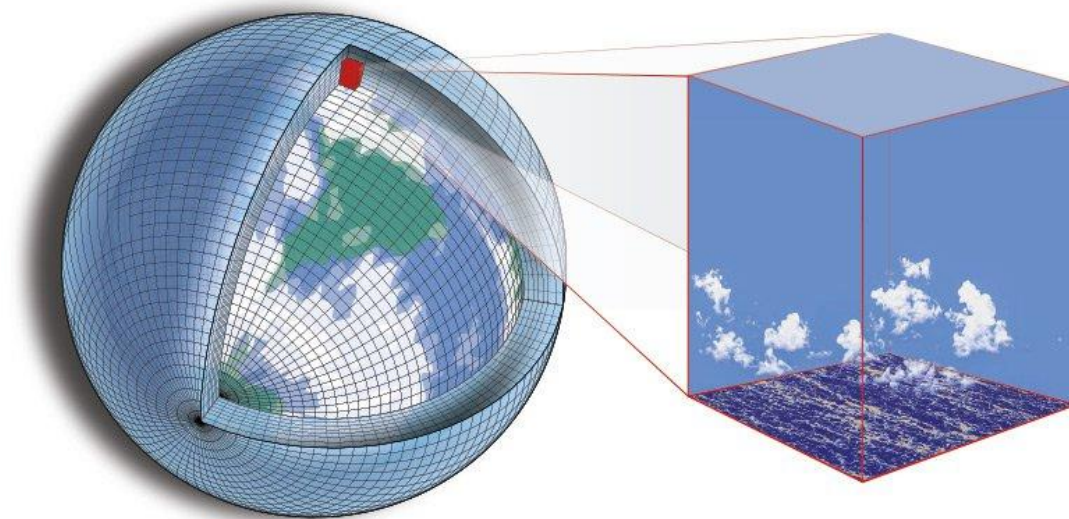
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What does interpolation do?

Providentia makes use of in-situ observations from specific measurement stations, at fixed locations.

When you have model data, its output is typically provided on a regular model grid at a fixed resolution.

Interpolation in Providentia refers to the mathematical process of estimating the model variable at the exact location of an observational station.

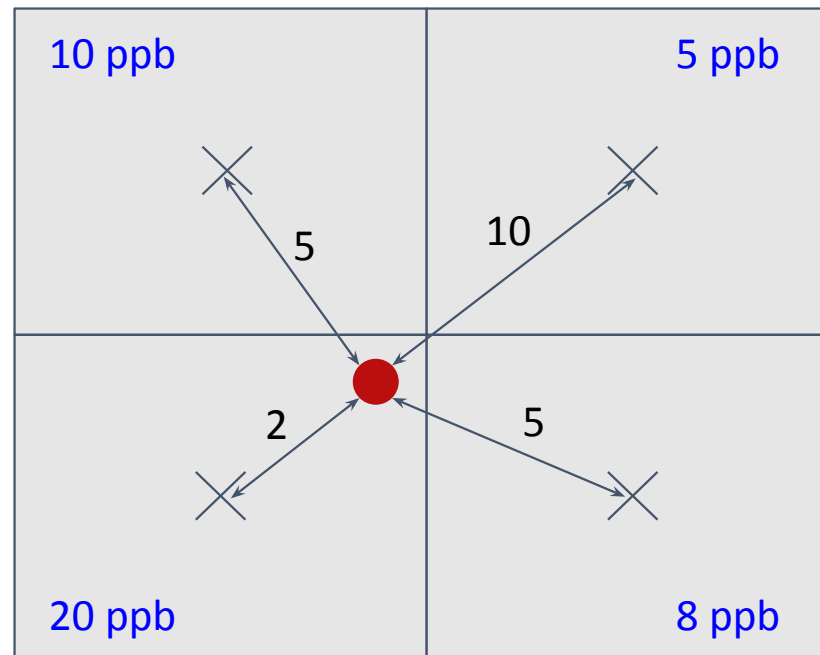


Inverse distance weighted interpolation

Inverse Distance Weighted interpolation estimates the value at an observational station by averaging model data in **N closest neighbour** grid cells, giving higher weight to grid cells that are closer.

N neighbours can be set via **interp_n_neighbours = N** in the configuration file. The default is **4**.

For 4 closest neighbours (default):



Weights

$$\frac{1}{2}, \frac{1}{5}, \frac{1}{5}, \frac{1}{10} \\ = 0.5, 0.2, 0.2, 0.1$$

Weighted average = **14.1 ppb**

Nearest neighbour = **20 ppb**

Mismatching temporal resolutions

Model simulations may have a different time resolution than observations and Providentia requires both temporal resolutions to match for evaluation. How Providentia handles this:

1. If model and observation resolutions already match → use data as is.
2. If the model resolution is finer (e.g. model: hourly, observations: 3 hourly) → **downsample** the model data.
3. If the model resolution is coarser (e.g. model: 3 hourly, observations: hourly) → **upsample** the model data.

In short, Providentia always adapts the model to the observations.

Mapping species

Sometimes the component provided in the observations does not exactly match the component given in a model. For example a model may provide information in 4D (e.g. time, lon, lat, alt), whereas in the observations it may be given in 3D (e.g. time, lon, lat). For these cases Providentia provides some allowance so that data can be mapped together.

This information can be set in **settings/mapping_species.yaml**, with the syntax for doing so being:

[observational component]

- **[model component]**

The typical use cases for mapping are:

1. 3D observational component to 4D model component. This can be either be through an additional vertical dimension in the model, e.g. sconco3 : vmro3 ; or an additional binned dimension in the model, e.g. vconcaerobin1 : loaddu
2. Observational component to fractional model component, e.g. pm10 : pm10du

settings/mapping_species.yaml

```
sconco3
- vmro3
vconcaerobin1
- loaddu
pm10
- pm10du
```

Run interpolation

If **--interp**, **--interpolate** or **--interpolation** is added as a launch option together with the mandatory configuration file on the command line, the interpolation will begin:

```
./bin/providentia --conf=configuration_name.conf --interp
```

```
[TRAINING]
network = nasa-aeronet/directsun_v3-lev15
species = od550aero
resolution = 3hourly
start_date = 20250701
end_date = 20260101
model = osuite-global-000, a00k_cirrus-regional-000
```

Note:

Each interpolation job covers one full month, thus for the given **start_date** and **end_date** only **YYYYMM** is considered, **DD** will be ignored.

Output Logs

When running the interpolation locally, information will be output in realtime to the command line as it goes on. Other logs will be generated in multiple different folders inside the ***logs/interpolation*** folder:

management_logs/ Informs on the output of the overall interpolation. Go here first.

interpolation_logs/ Stores output information of individual interpolations.

For more information about the interpolation, check the official documentation:

<https://providentia.readthedocs.io/en/latest/Interpolation.html>

Dashboard



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Dashboard

Providentia

— □ ×

Data Selection

γ_v3-lev15

3hourly

aod

20250701

od550aero

20260101

QA

FLAGS

MODELS

SPECIES

READ

Filters

Bounds

0.0

1

COVERAGE

PERIOD

RESET

METADATA

FILTER

Statistics

Mode

Temporal

Aggregation

Median

Colocation

Temporal

Resampling

None

Site Selection

☒ All

☐ Intersect

☐ Extent

↓

📄

🌐

🔍

←

→

📏

🔄

🗺️

167 Selected Stations of 167 Available

0.0168 0.1024 0.1881 0.2737 0.3594 0.4450 0.5307

Mean [unitless]

statsumme distribution

	Mean	StdDev	p5	Median	p95
observations	0.13	0.09	0.04	0.09	0.32
osuite-global-000	0.01	0.03	2.79e-04	0.00	0.06
a00k_cirrus-regional-000	0.03	0.05	4.24e-08	0.00	0.14

Density

0.00 0.05 0.10 0.15 0.20 0.25 0.30

[unitless]

● observations ● osuite-global-000 ● a00k_cirrus-regional-000

timeseries

[unitless]

Time

2025-08-01 2025-09-01 2025-10-01 2025-11-01 2025-12-01 2026-01-01

periodic

Mean [unitless]

0 3 6 9 12 15 18

A S O N D

M T W T F S S

Main menu

Data Selection					Filters			Statistics		Colocation	Resampling	Site Selection	
r_v3-lev15	aod	od550aero	QA	MODELS	Bounds	0.0	1	Mode	Temporal S	☒ Temporal	None	☒ All	
3hourly	20250701	20260101	FLAGS	SPECIES	COVERAGE	PERIOD	METADATA	Aggregation	Median			☐ Intersect	
READ					RESET			FILTER					☐ Extent
													

Data selection menu

Network	Matrix	Species	Quality assurance (GHOST)	Models
n_v3-lev15	aod	od550aero	QA	MODELS

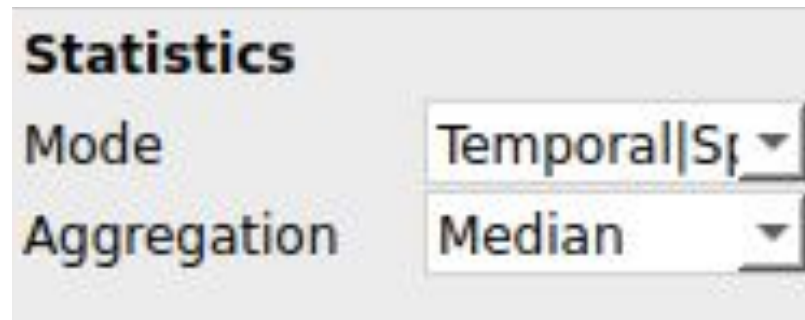
Temporal resolution	Start date	End date	Data flags (Provider)	Multispecies filtering
3hourly	20250701	20260101	FLAGS	SPECIES

Filters menu

	Data lower bound	Data upper bound
Bounds	0.0	1

Representativity filters	Time period filters	Metadata filters
% REP	PERIOD	META

Statistics menu



Statistic mode

Statistic aggregation

Colocation menu



Colocate models vs. observations, removing temporal gaps

Resampling menu



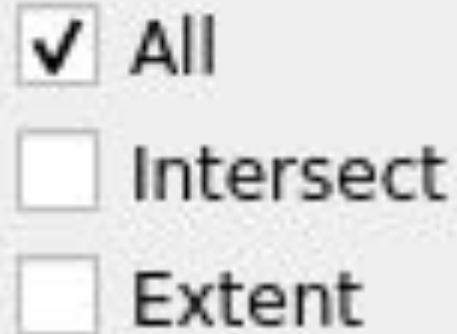
Temporal resolution to resample your data to (always lower than the selected resolution)

Stations selection menu

Select all stations

Select intersecting stations
within all model domains

Select stations on current map view



A screenshot of a software menu for selecting stations. It contains three items, each with a checkbox on the left and a label on the right. The first item, 'All', has a checked checkbox. The second item, 'Intersect', has an unchecked checkbox. The third item, 'Extent', has an unchecked checkbox.

☒	All
☐	Intersect
☐	Extent

Icons menu

Upload
configuration files

Set original
view

Forward

Pan

Save
canvas



Download data and
configuration files

Set world
view

Back

Zoom to
rectangle

Lasso

Interactive features



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Navigation tips

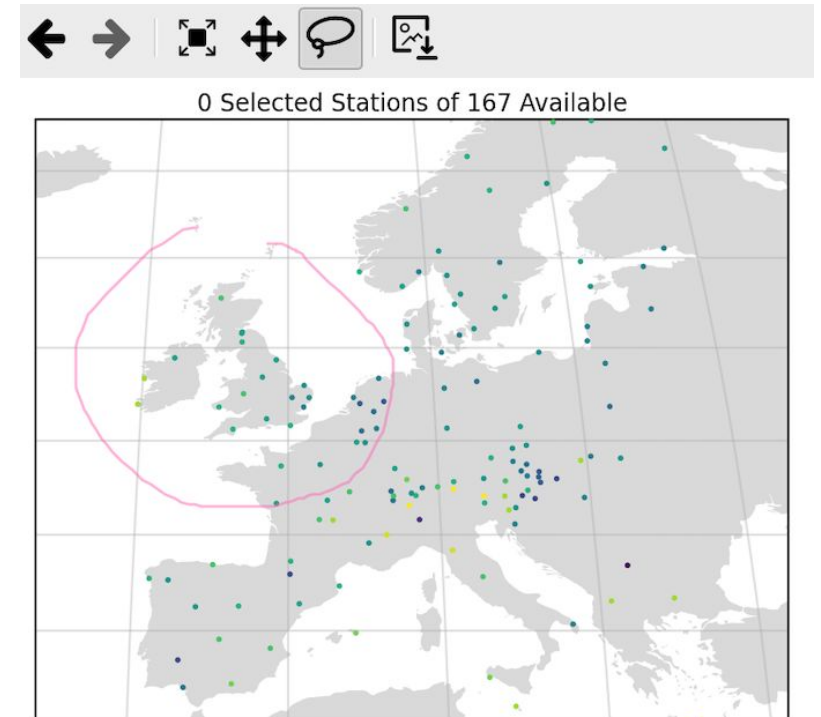
Individual stations can be selected by left clicking on them on the map.

Multiple stations can be selected by activating the lasso, and drawing it round all desired stations.

To unselect stations simply click on an empty space on the map.

When multiple stations are selected, to add or remove any individual station right click on the given station.

To quickly zoom on the map, the mouse or trackpad scroll wheel can be used.



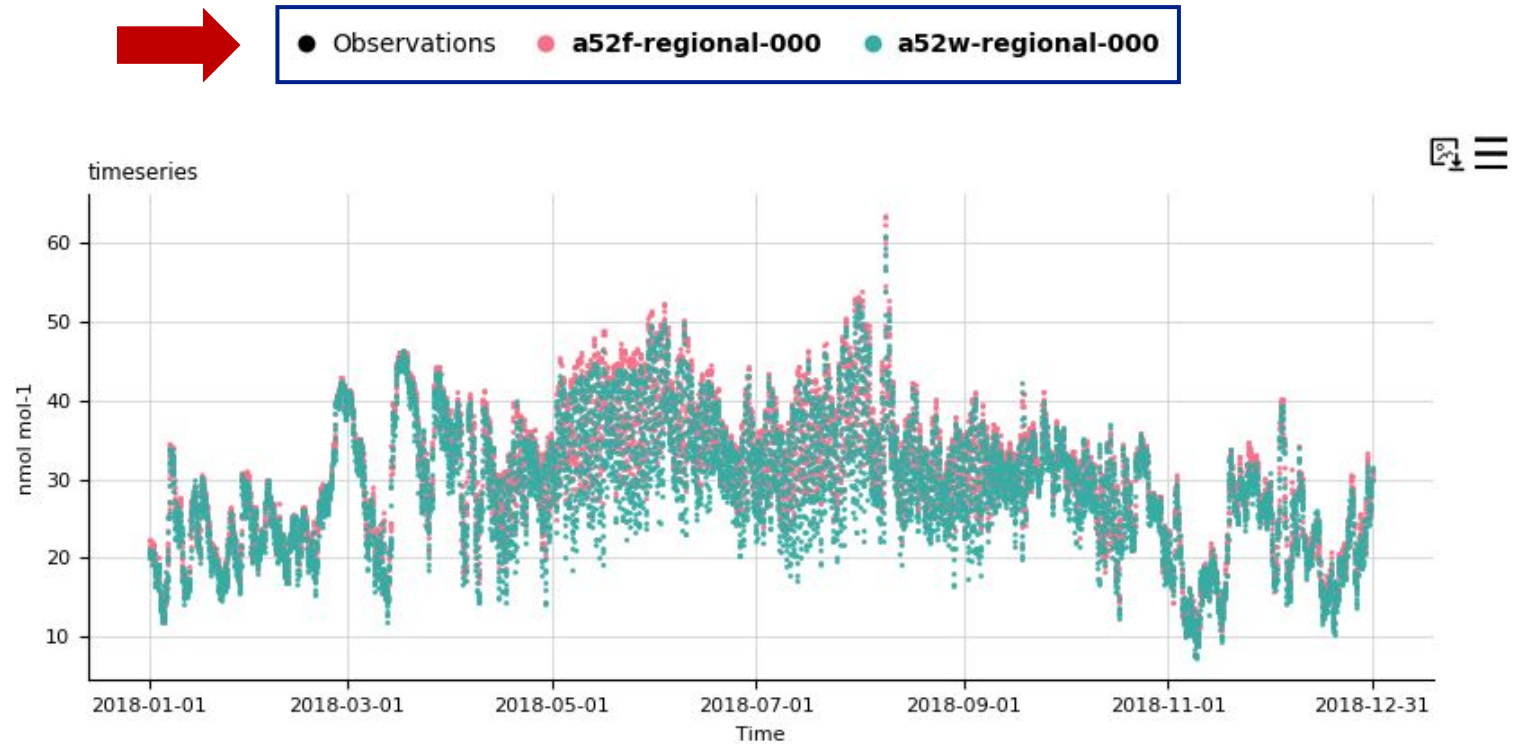
Example of lasso selection

Legend picking

Clicking on the legend labels will remove or add data to each of the plots

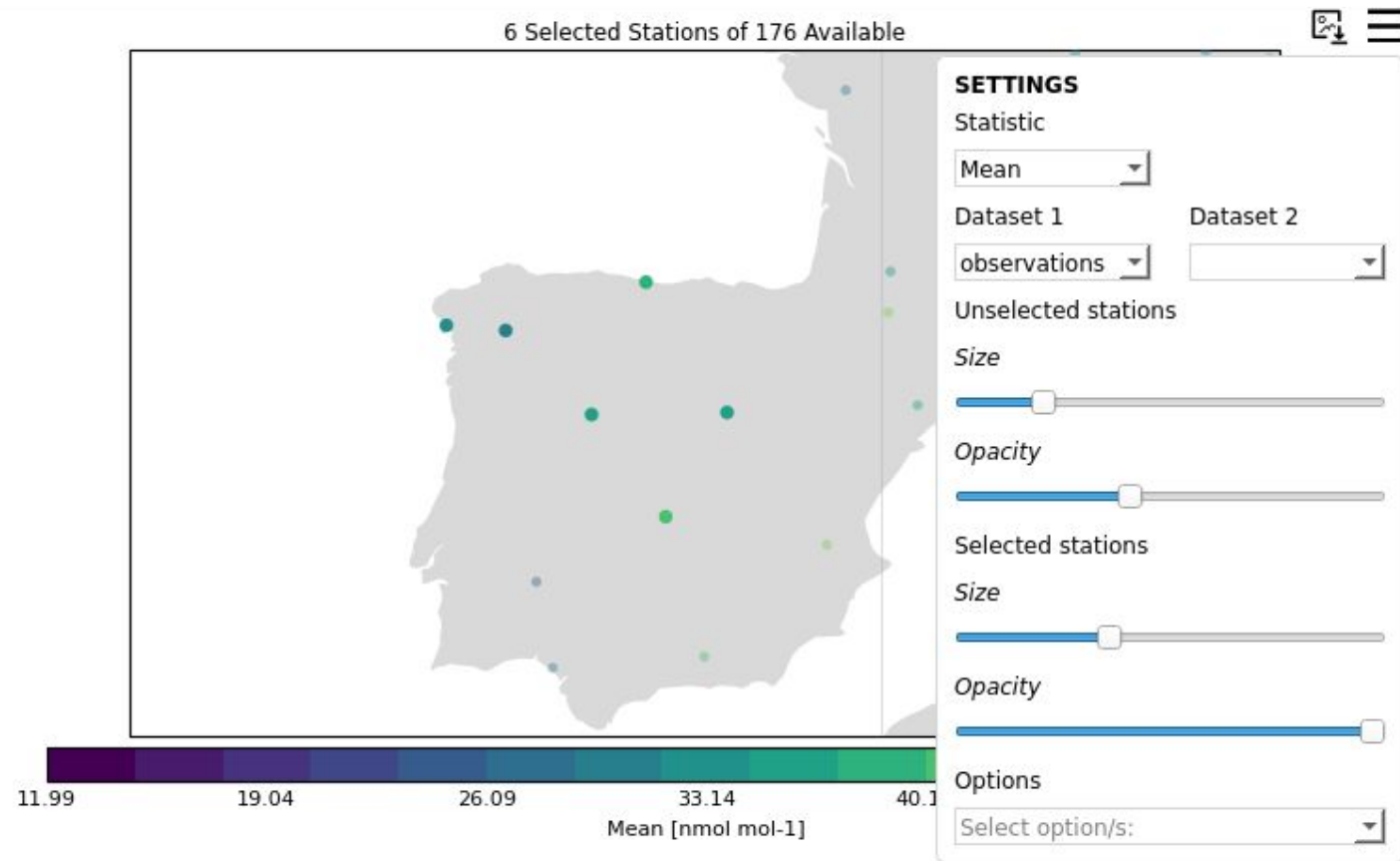
Bold = Visible

Roman = Invisible



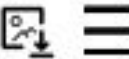
Customise the plot style

The style of the plots can be edited by clicking on the **burger menus** and changing the settings.



Choosing the statistics

The statistics in the statsummary can be updated from the burger menu.

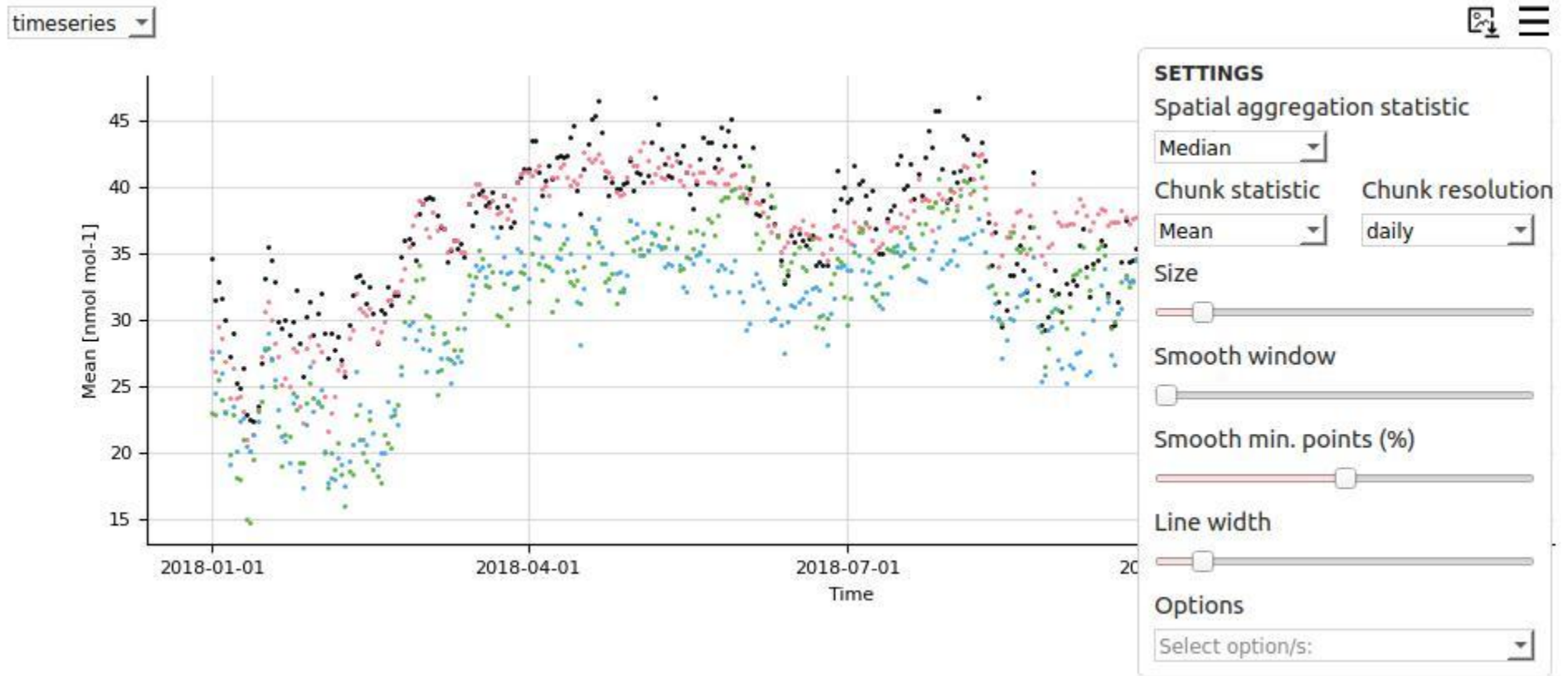
statsummary 

SETTINGS
Periodic cycle: None
Statistic: , Median, p95
Periodic mode: Independent
Periodic aggregation: Mean
Options: Select option/s:

	Mean	StdDev	p5				
Observations	34.01	13.98	10.96				
cams61_camsra_ph2-eu-000	30.21	13.27	7.40				
cams61_chimere_ph2-eu-000	36.24	9.37	20.07	36.76	50.57	100.98	79.17
cams61_dehm_ph2-eu-000	42.40	10.27	24.92	42.74	58.55	117.41	79.17

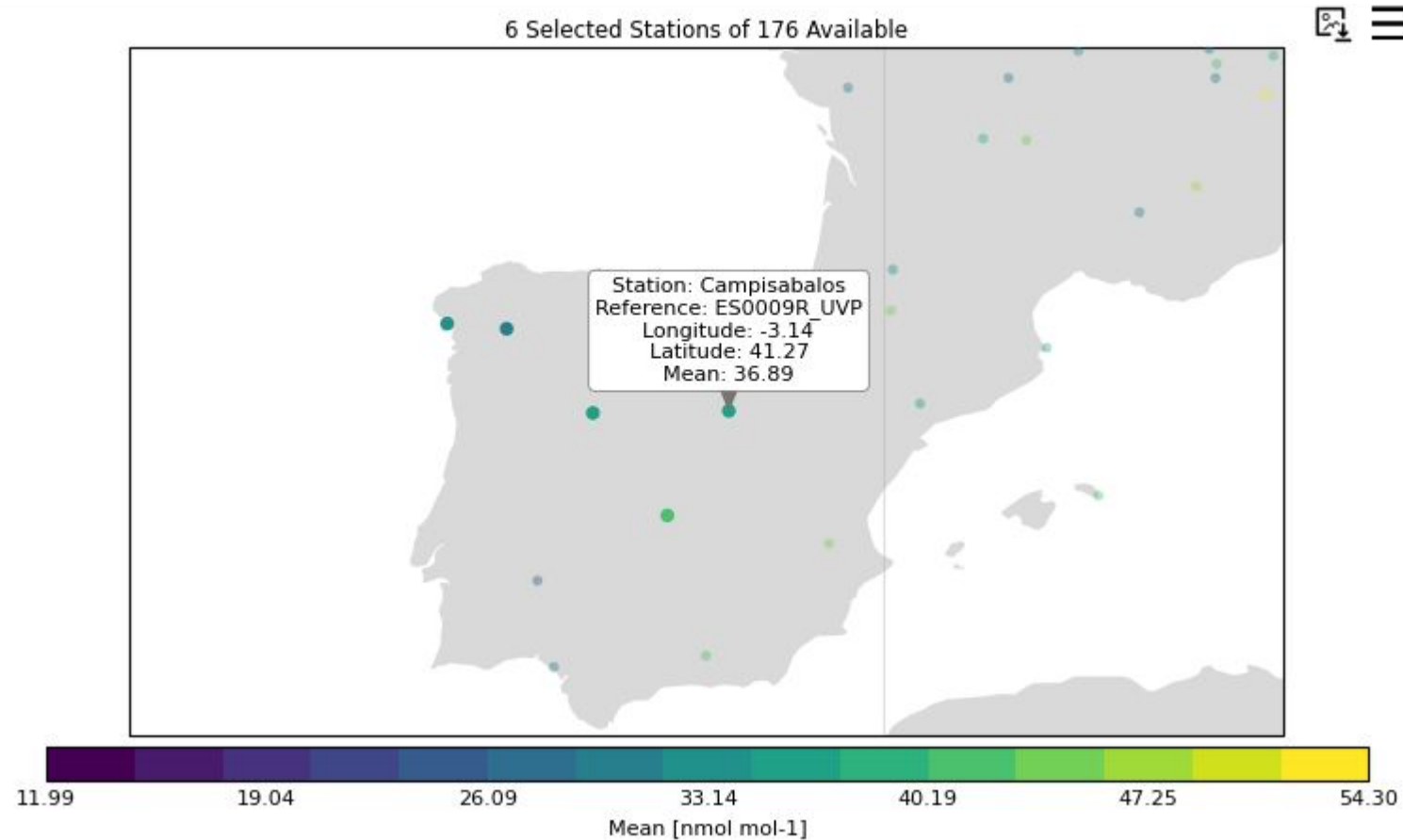
Choosing the statistics

The statistics in other plots can also be updated from the burger menus.



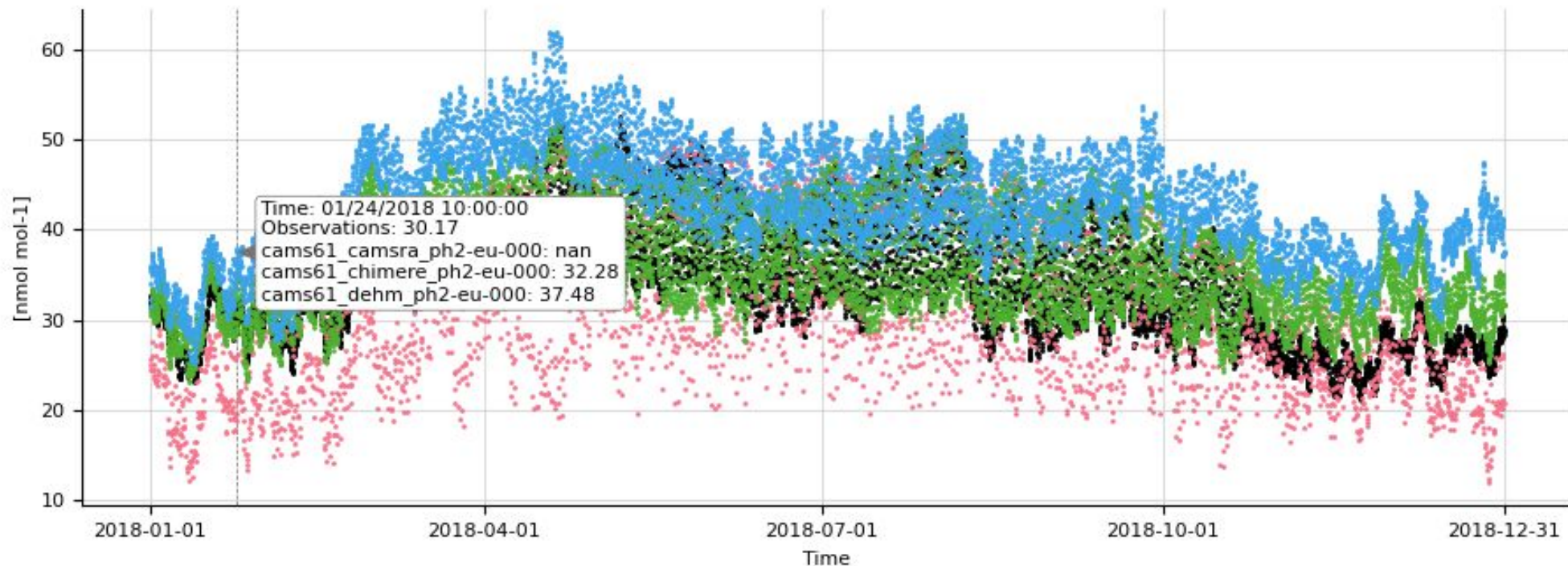
Information on hover

We can see the stations details and data by hovering on the map.



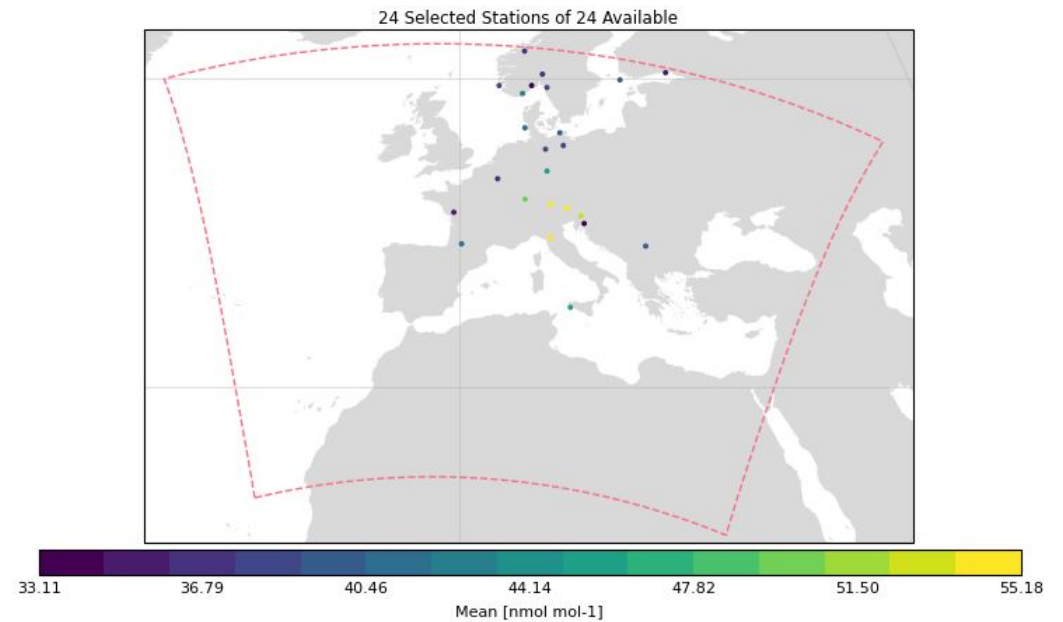
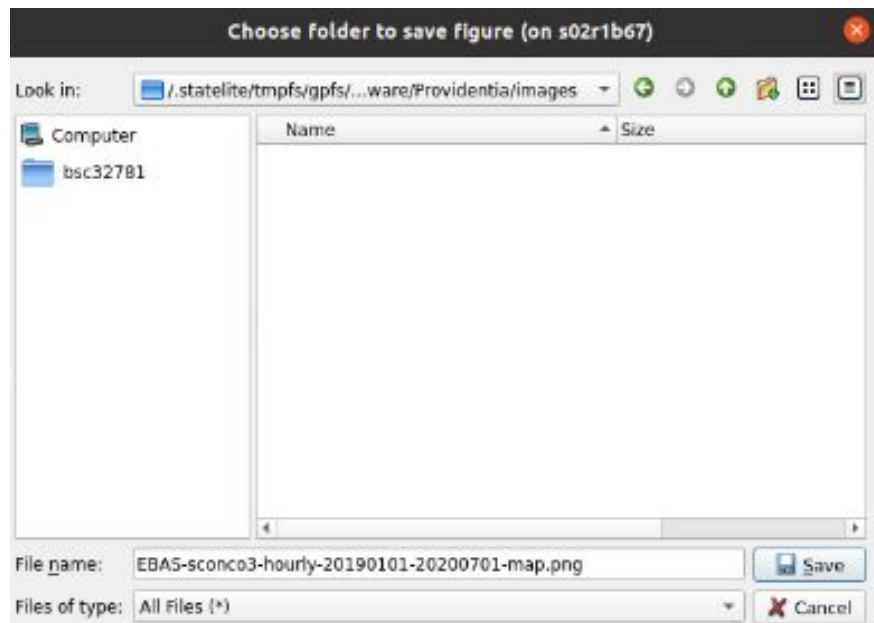
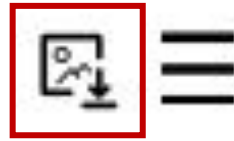
Information on hover

We can also check the values of each dataset by hovering on the other plot types.



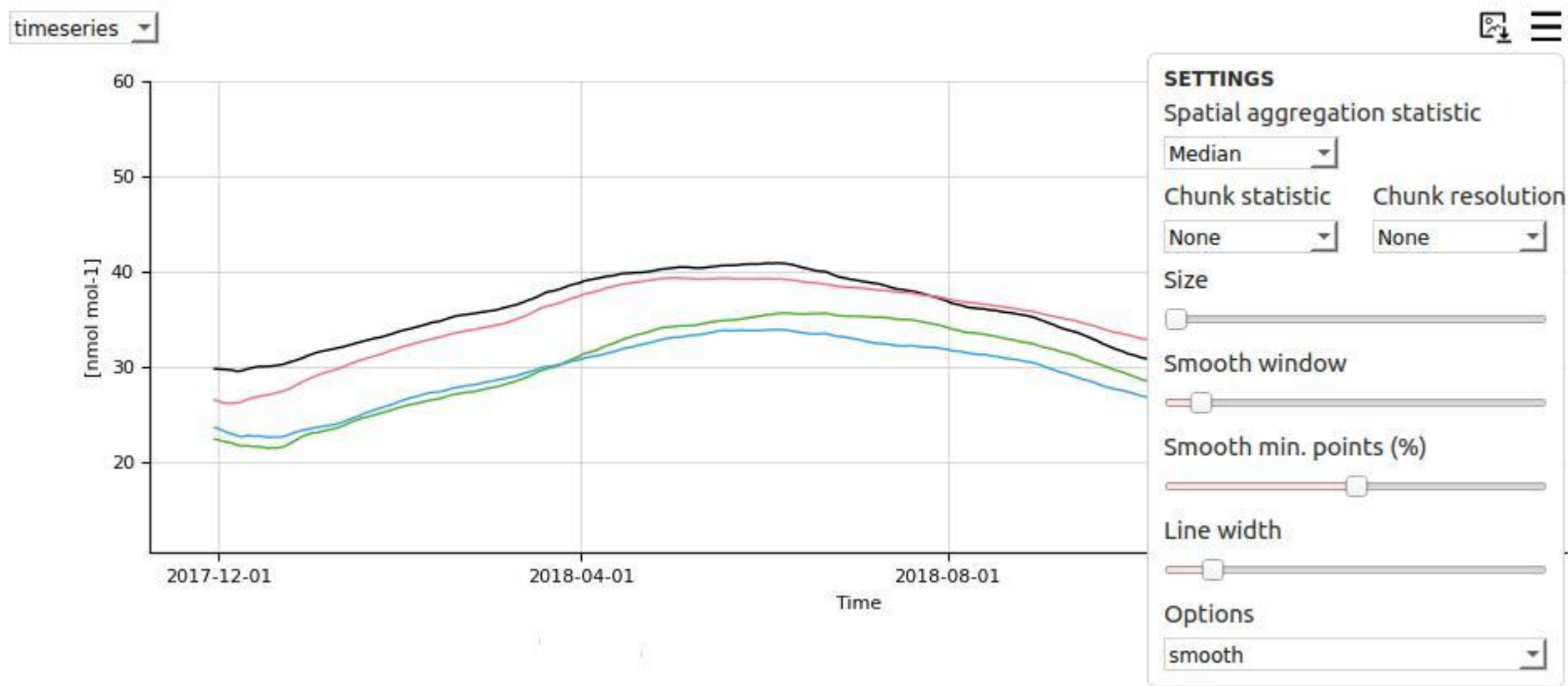
Saving plot figures

Saving plot figures is now possible by clicking on the image icons next to the burger menus



Smoothing

It is possible to add a smooth line in the timeseries plot by editing the smooth window, the data points can be then hidden by reducing their size to 0.



BREAK
20 min



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Colocation



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Spatial colocation

When **loading more than one species** (e.g. when using multispecies filtering or in multispecies plots in the report and library modes) you may want to ensure that the available stations **measure data for all** species that are to be loaded. To do this, we need to activate spatial colocation.

After activating spatial colocation, any stations that do not have valid data for any of the loaded species are dropped.

Spatial colocation can be set in the configuration file by setting a boolean as follows (default: True):

```
spatial_colocation = False
```

Reference:

<https://providentia.readthedocs.io/en/latest/Colocation.html>

Ozone: 1731 stations

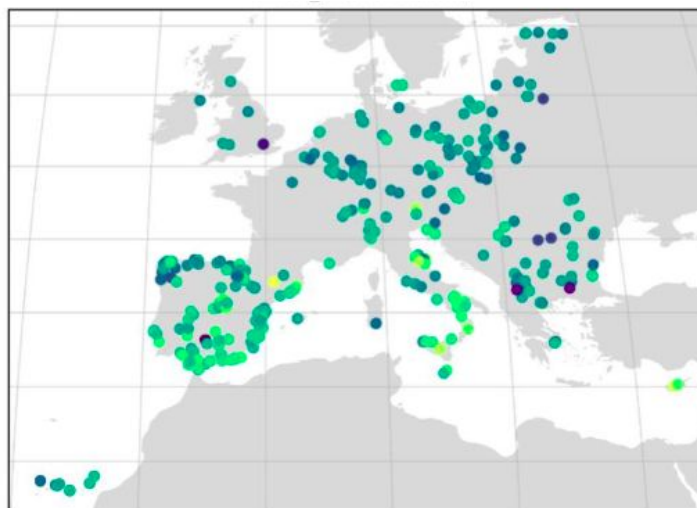


CO: 635 stations



Without spatial colocation

Ozone: 366 stations



CO: 366 stations



With spatial colocation

Temporal colocation

Temporal colocation is used to **temporally pair observations and model data**, with any missing measurements in either the observational or model array, imposing missing measurements on the other.

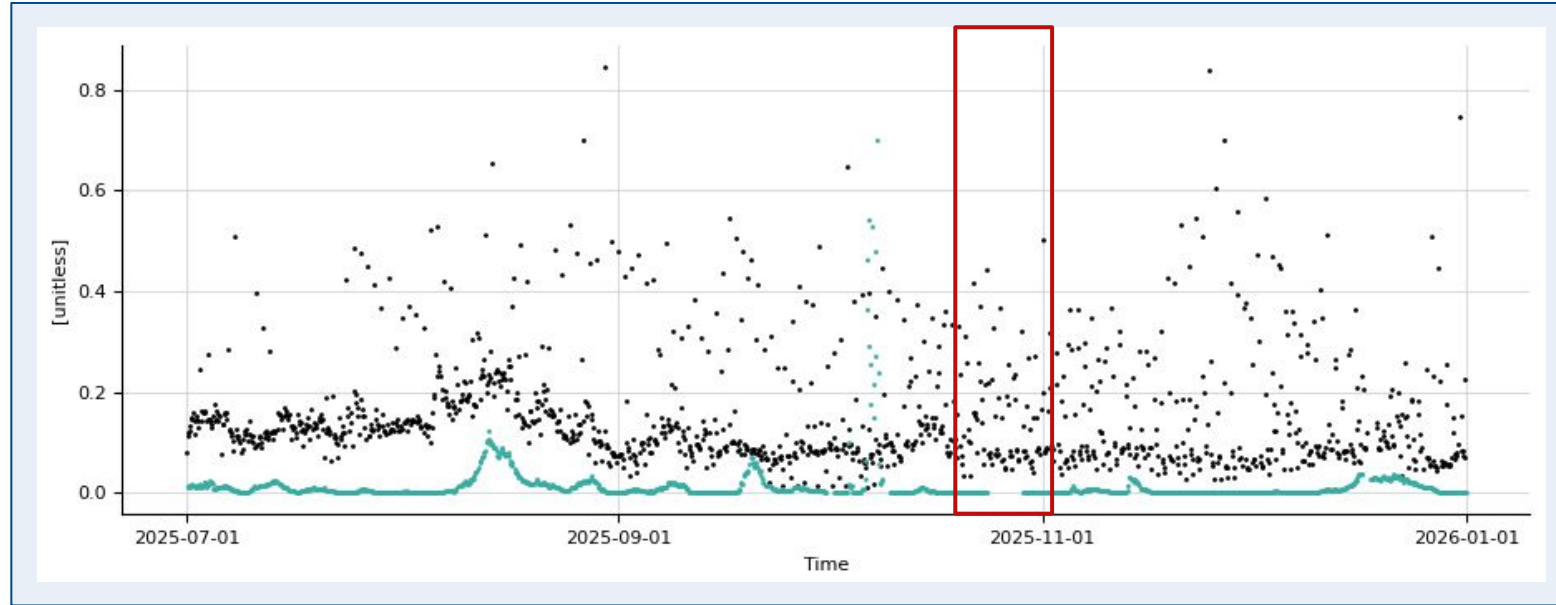
When temporal colocation is active, you will have access to more plot types (scatter, taylor, fairmode-target, and fairmode-statsummary).

Temporal colocation can be set in the configuration file by setting a boolean as follows (default: True):

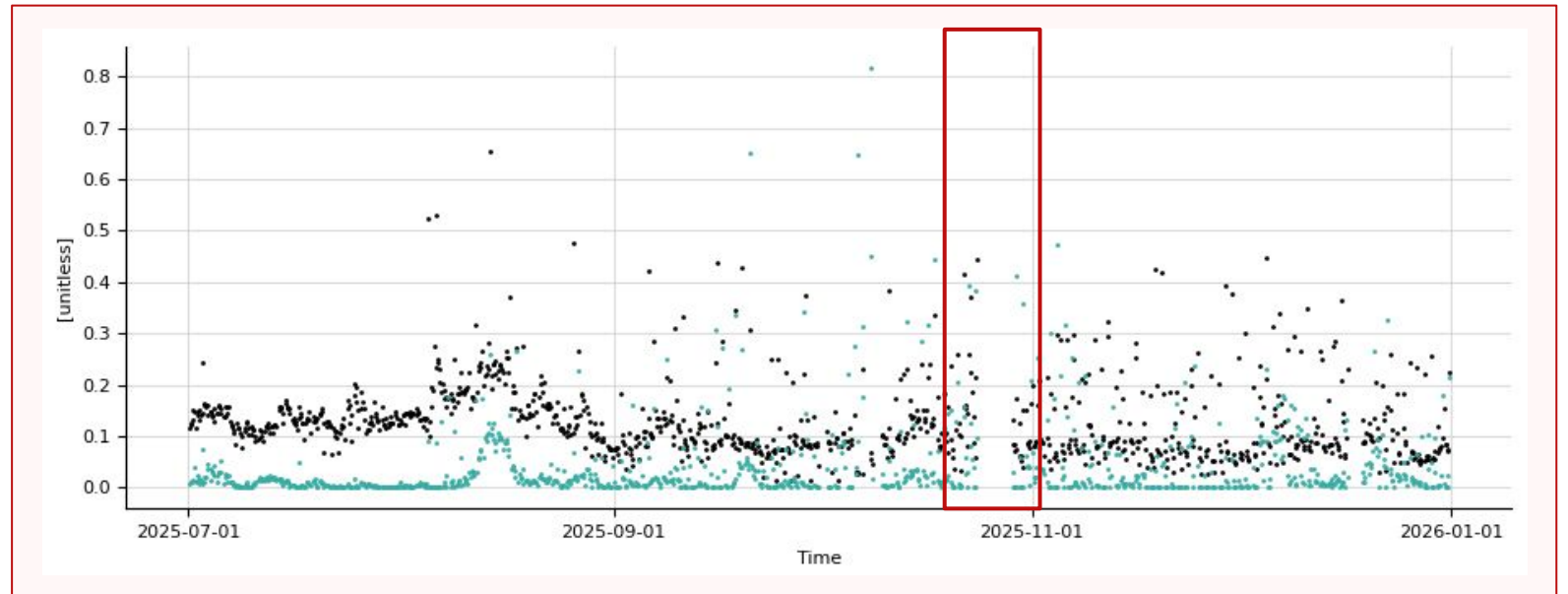
```
temporal_colocation = False
```

Reference: <https://providentia.readthedocs.io/en/latest/Colocation.html>

Without temporal colocation



With temporal colocation



Filtering



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Metadata filtering

The most common type of data filtering is by metadata, e.g. for **country**, **longitude**, **altitude** etc.

Metadata can either be **numerical** or **text** and the method for filtering for these slightly varies, as will be described.

One common question is what metadata available variables exist for the loaded data that can be filtered by. These can be seen empirically by looking through the sub-menus under the **META** button on the dashboard, with all available variables are organised into 5 categories: **station position**, **station classifications**, **station miscellaneous**, **globally gridded classifications** and **measurement process information**.

Metadata filtering: Numerical data

Dashboard

The dashboard interface for metadata filtering includes the following sections:

- Filters**: A section with input fields for 'Bounds' (0.0 and 1) and buttons for 'COVERAGE', 'PERIOD', 'METADATA' (circled in red), 'RESET', and 'FILTER' (circled in red).
- Select metadata type to filter stations by**: A list of metadata types where 'STATION POSITION' is circled in red.
- Filter stations by measurement position**: A section with 'HOME' and 'PREVIOUS' buttons, and a table for filtering by measurement position.
- Filters**: A second section with input fields for 'Bounds' (0.0 and 20.0) and buttons for '% REP', 'PERIOD', 'META', 'RESET', and 'FILTER' (circled in red).

	Min	Max	A
latitude	30	72	☒
longitude	511465454	870803833	☐
altitude	1.0	3578.0	☐
sampling_height	0.0	50.0	☐
measurement_altitude	3.0	3578.0	☐

Configuration file

latitude = 30, 72

Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#numerical-metadata>

Metadata filtering: Text metadata

Dashboard

Configuration file

1

Filters

Bounds 0.0 1

COVERAGE PERIOD **METADATA**

RESET FILTER

2

Select metadata type to filter stations by

STATION POSITION
STATION CLASSIFICATIONS
STATION MISCELLANEOUS
GLOBALLY GRIDDED CLASSIFICATIONS
MEASUREMENT PROCESS INFORMATION

station_reference = Kuwait_University_2

3

Filter stations by unique station_reference

HOME PREVIOUS

	K	R		K	R		K	R
AAU_Jackros_ET	☐	☐	Dunkerque	☐	☐	Kaopio	☐	☐
ALERIA_MF	☐	☐	Dushanbe	☐	☐	Kuwait_University_2	☒	☐
ARM_Graciosa	☐	☐	ESP-VLC-ELPALMAR	☐	☐	LAQTO-STATION	☐	☐
ATHENS_NTUA	☐	☐	El_Arenosillo	☐	☐	LAQUILA_Coppito	☐	☐
Agen_Palissy	☐	☐	Ersa	☐	☐	Lahore	☐	☐
AgiaGalini_AUTH	☐	☐	Evora	☐	☐	Lampedusa	☐	☐
AgiaMarina_Xyliatou	☐	☐	Exeter_MO	☐	☐	Lecce_University	☐	☐
Aire_sur_Adour_MF	☐	☐	FORTH_RSLab	☐	☐	Leipzig	☐	☐
Amity_Univ_Gurgaon	☐	☐	FZJ-JOYCE	☐	☐	Lerwick_MO	☐	☐
Andenes	☐	☐	Finokalia-FKL	☐	☐	Lille	☐	☐
Antikythera_NOA	☐	☐	Garmisch_KIT	☐	☐	Loftus_MO	☐	☐
Aosta	☐	☐	Glasgow_MO	☐	☐	MAGIC ORM_La_Palma	☐	☐
Apulian_Tavoliere	☐	☐	Gozo	☐	☐	Madrid	☐	☐
Aras_de_los_Olmos	☐	☐	Granada	☐	☐	Magurele_Inoe	☐	☐
Arcachon	☐	☐	Greifswald_Univ	☐	☐	Mainz	☐	☐
Aubiere_LAMP	☐	☐	Guadiana_UGR_4	☐	☐	Majadas	☐	☐
Badajoz	☐	☐	Hanle_Observatory	☐	☐	Manchester_UoM	☐	☐

4

Filters

Bounds 0.0 1

COVERAGE PERIOD **METADATA**

RESET **FILTER**

Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#text-metadata>

Multispecies filtering

Multispecies filtering refers to the ability to **filter loaded species data by the values of another species**.

For example, when performing investigations of dust in the atmosphere it is a common practice to filter the AOD by the Angstrom exponent, to isolate values associated with dust.

Dashboard

Data Selection

r_v3-lev15	aod	od550aero	QA	MODELS
3hourly	20250701	20260101	FLAGS	SPECIES
READ				

Select Network/s and Specie/s to Filter by

Network	Matrix	Species	Min	Max	Fill value	A
T_v3-lev1.5	aod	140-870aero	>0.6	:	nan	☒

Data Selection

r_v3-lev15	aod	od550aero	QA	MODELS
3hourly	20250701	20260101	FLAGS	SPECIES
READ				

Configuration file

```
network = AERONET_v3_lev1.5
species = od550aero
filter_species = AERONET_v3_lev1.5:ae440-870aero (>0.6, :, nan)
spatial_colocation = True
```

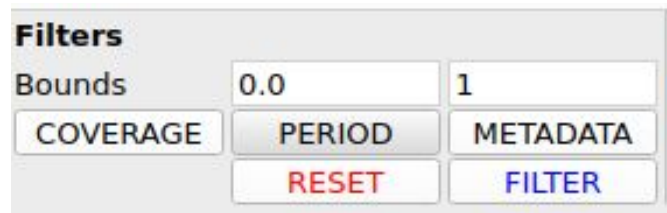
Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#multispecies-filtering>

Bounds

Often it is desired to remove values which exceed certain extreme bounds, as it is known that data should appear at such extremes. These bounds will be **by default active in Providentia**, with extreme bounds associated with a given species taken from definitions in GHOST.

Dashboard



Filters

Bounds

```
'extreme_lower_limit':0.0, 'extreme_upper_limit':400.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':1200.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':600.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':3000.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':30000.0,  
'extreme_lower_limit':0.0, 'extreme_upper_limit':50000.0,
```

Configuration file

```
lower_bound = 0  
upper_bound = 1
```

Link to [GHOST Standards](https://ghoststandards.org/)

Reference:

<https://providentia.readthedocs.io/en/latest/Filtering.html#bounds>

QA and flags in GHOST



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GHOST QA

GHOST QA refers to quality assurance flags which come from checks designed in GHOST to inform on the quality of data, and can be used to filter measurements.

These can be set via the configuration file either by code or name as follows:

qa = 0, 1, 2

qa = Missing Measurement, Infinite Value, Negative Measurement

QA flags can also be added to, or subtracted from as follows (again, either code or name):

add_qa = Missing Measurement, Negative Measurement

subtract_qa = 0, 2

On the dashboard the GHOST QA flags can be set interactively via the **QA** button.

See [here](#) for specific GHOST QA flag definitions and codes.

QA flags

Select standardised quality assurance flags to filter by

Select AllClear AllSelect Default

Missing Measurement	☒	Above Preferential Upper Limit of Detection	☒
Infinite Value	☒	Insufficient Measurement Resolution - Documented	☐
Negative Measurement	☒	Insufficient Measurement Resolution - Reported	☐
Zero Measurement	☐	Insufficient Measurement Resolution - Preferential	☒
Not Maximum Data Quality Level	☐	Insufficient Measurement Resolution - Empirical	☒
Preliminary Data	☐	Persistent Recurring Values - 5/6	☐
Invalid Data Provider Flags - GHOST Decreased	☒	Persistent Recurring Values - 9/12	☐
Invalid Data Provider Flags - Network Decreased	☐	Persistent Recurring Values - 16/24	☐
No Valid Data to Average	☒	Monthly Fractional Unique Values <= 1%	☐
Methodology Not Mapped	☐	Monthly Fractional Unique Values <= 5%	☐
Assumed Primary Sampling	☐	Monthly Fractional Unique Values <= 10%	☐
Assumed Sample Preparation	☐	Monthly Fractional Unique Values <= 30%	☐
Assumed Measurement Methodology	☐	Monthly Fractional Unique Values <= 50%	☐
Unknown Primary Sampling Type	☐	Monthly Fractional Unique Values <= 70%	☐
Unknown Primary Sampling Instrument	☐	Monthly Fractional Unique Values <= 90%	☐
Unknown Sample Preparation Type	☐	Data Outlier - Exceeds Scientifically Decreased Lower/Upper Limit	☒
Unknown Sample Preparation Technique	☐	Data Outlier - Monthly Median Exceeds Scientifically Decreased Upper Limit	☒
Unknown Measurement Method	☐	Data Outlier - Network Decreased	☒
Unknown Measuring Instrument	☐	Data Outlier - Manually Decreased	☒
Erroneous Primary Sampling	☒	Possible Data Outlier - Monthly Adjusted Boxplot	☐
Erroneous Sample Preparation	☒	Probable Data Outlier - Monthly Adjusted Boxplot	☒
Erroneous Measurement Methodology	☒	Monthly Distribution Consistency - Zone 1	☐
Invalid QA Measurement Methodology	☐	Monthly Distribution Consistency - Zone 2	☐
Corrected Parameter	☐	Monthly Distribution Consistency - Zone 3	☐
Sample Gas Volume - Network Standard	☐	Monthly Distribution Consistency - Zone 4	☐
Sample Gas Volume - Unknown	☐	Monthly Distribution Consistency - Zone 5	☐
Unit Conversion - Network Standard Sample Gas Volume Assumption	☐	Monthly Distribution Consistency - Zone 6	☐
Unit Conversion - Educated Guess Sample Gas Volume Assumption	☐	Monthly Distribution Consistency - Zone 7	☐
Station Position Doubt - DEM Decreased	☐	Monthly Distribution Consistency - Zone 8	☐
Station Position Doubt - Manually Decreased	☐	Monthly Distribution Consistency - Zone 9	☐
Data Product	☐	Monthly Distribution Consistency - Zone 10	☐
Insufficient Data to Calculate Data Product	☐	Monthly Distribution Consistency - Unclassified	☐
Local Precipitation	☐	Systematic Inconsistent Monthly Distributions - 2/3 Months >= Zone 6	☐
Local Extreme Weather	☐	Systematic Inconsistent Monthly Distributions - 4/6 Months >= Zone 6	☒
Local Atmospheric Obscuration	☐	Systematic Inconsistent Monthly Distributions - 8/12 Months >= Zone 6	☒
Local Contamination	☐		
Local Exceptional Event	☐		
Non-Integer Local Timezone (relative to UTC)	☐		
Timezone Doubt	☐		
Below Documented Lower Limit of Detection	☐		
Below Reported Lower Limit of Detection	☐		
Below Preferential Lower Limit of Detection	☒		
Above Documented Upper Limit of Detection	☐		
Above Reported Upper Limit of Detection	☐		

Network flags

Network QA refers to standardised quality assurance flags which come from networks processed in GHOST which inform on the quality of data, and can be used to filter measurements.

These can be set via the configuration file either by code or name as follows:

flags = 0, 1, 2

flags = Valid Data, Preliminary Data, Missing Data

Network flags can also be added to, or subtracted from as follows (again, either code or name):

add_flags = Valid Data, Missing Data

subtract_flags = 0, 2

On the dashboard the network flags can be set interactively via the **FLAGS** button.

See [here](#) for specific Network QA flag definitions and codes.

Network flags

Select standardised data reporter provided flags to filter by

Select All Clear All Select Default

☐ Valid Data	☐ Calibration	☐ Freezing Rain - Moderate	☐ Diamond Out - Moderate	☐ Volcanic Eruptions
☐ Preliminary Data	☐ Accuracy Check	☐ Freezing Rain - Heavy	☐ Diamond Out - Heavy	☐ Wildfire
☐ Missing Data	☐ Blank Check	☐ Freezing Rain Showers - Unspecified Intensity	☐ Glass	☐ Chemical/Industrial Accident
☐ Invalid Data - Unspecified	☐ Detection Limits Check	☐ Freezing Rain Showers - Light	☐ Haze	☐ Cleanup After a Major Disaster
☐ Un-Flagged Data	☐ Precision Check	☐ Freezing Rain Showers - Moderate	☐ Thunderstorm	☐ Demolition
☐ Estimated Data - Unspecified	☐ Response Time Check	☐ Freezing Rain Showers - Heavy	☐ Funnel Clouds	☐ Fireworks
☐ Estimated Data - Measured Negative Value	☐ Span Check	☐ Snow - Unspecified Intensity	☐ Squalls	☐ Infrequent Large Gathering
☐ Estimated Data - No Value Detected	☐ Zero Check	☐ Snow - Light	☐ Tropical Cyclone (Cyclone/Hurricane/Typhoon)	☐ Tornado Act
☐ Estimated Data - Value Below Detection Limit	☐ Instrumental Inspection	☐ Snow - Moderate	☐ Duststorm	☐ Visibility Distance Unlimited
☐ Estimated Data - Value Above Detection Limit	☐ Instrumental Repair	☐ Snow - Heavy	☐ Sandstorm	☐ Ceiling Height Unlimited
☐ Estimated Data - Value Substituted From Secondary Monitor	☐ Quality Control Audit	☐ Snow Showers - Unspecified Intensity	☐ Dustified Whirls	☐ AERONET L2 ADD
☐ Estimated Data - Multiple Parameters Aggregated	☐ Data Formatting/Processing Issue	☐ Snow Showers - Light	☐ High Winds	☐ AERONET L2 WITHOUT ADD
☐ Estimated/Regular Data - Unspecified	☐ Connected Data Formatting/Processing Issue	☐ Snow Showers - Moderate	☐ No Atmospheric Observation	
	☐ Aggregation/Representation Issue - Unspecified	☐ Snow Showers - Heavy	☐ Atmospheric Obscuration - Unknown	
☐ Data Does Not Meet Internal Network Quality Control Criteria	☐ Data Window Completeness < 50%	☐ Hail - Unspecified Intensity	☐ Dust	
☐ High Variability of Data	☐ Data Window Completeness < 75%	☐ Hail - Light	☐ Blowing Dust	
☐ Irregular Data Manually Screened and Accepted	☐ Data Window Completeness < 85%	☐ Hail - Moderate	☐ Drifting Dust	
☐ Negative Value	☐ Data Window Completeness < 90%	☐ Hail - Heavy	☐ Sand	
☐ No Value Detected	☐ Data Window Completeness < 95%	☐ Hail Showers - Unspecified Intensity	☐ Blowing Sand	
☐ Reasonable/Recalculated Data	☐ >= 75% of Measurements in Window Below Detection Limit	☐ Hail Showers - Light	☐ Drifting Sand	
☐ Value Close to Detection Limit	☐ >= 50% of Measurements in Window Below Detection Limit	☐ Hail Showers - Moderate	☐ Blowing Snow	
☐ Value Below Acceptable Range	☐ No Significant Weather	☐ Hail Showers - Heavy	☐ Drifting Snow	
☐ Value Above Acceptable Range	☐ Precipitation - Unspecified Intensity	☐ Ice Pellets - Unspecified Intensity	☐ Fog	
☐ Value Below Detection Limit	☐ Precipitation - Light	☐ Ice Pellets - Light	☐ Freezing Fog	
☐ Value Above Detection Limit	☐ Precipitation - Moderate	☐ Ice Pellets - Moderate	☐ Ground Fog	
☐ Measurement Issue - Unspecified	☐ Precipitation - Heavy	☐ Ice Pellets - Heavy	☐ Ice Fog	
☐ Chemical Issue	☐ Drizzle - Unspecified Intensity	☐ Ice Pellets Showers - Unspecified Intensity	☐ Haze	
☐ Erroneous Sampling Operation	☐ Drizzle - Light	☐ Ice Pellets Showers - Light	☐ Mist	
☐ Extreme Internal Instrument Meteorological Conditions	☐ Drizzle - Moderate	☐ Ice Pellets Showers - Moderate	☐ Sea Spray	
☐ Extreme Ambient Laboratory Meteorological Conditions	☐ Drizzle - Heavy	☐ Ice Pellets Showers - Heavy	☐ Smoke	
☐ Extreme External Meteorological Conditions	☐ Freezing Drizzle - Unspecified Intensity	☐ Snow Pellets - Unspecified Intensity	☐ Volcanic Ash	
☐ Extreme Sample Transport Conditions	☐ Freezing Drizzle - Light	☐ Snow Pellets - Light	☐ No Local Contamination	
☐ Invalid Flow Rate	☐ Freezing Drizzle - Moderate	☐ Snow Pellets - Moderate	☐ Local Contamination - Unspecified	
☐ Human Error	☐ Freezing Drizzle - Heavy	☐ Snow Pellets - Heavy	☐ Agricultural Contamination	
☐ Matrix Effect	☐ Rain - Unspecified Intensity	☐ Snow Pellets Showers - Unspecified Intensity	☐ Bird-Dropping Contamination	
☐ Mechanical Issue/Non-Operational Equipment	☐ Rain - Light	☐ Snow Pellets Showers - Light	☐ Construction Contamination	
☐ No Technician	☐ Rain - Moderate	☐ Snow Pellets Showers - Moderate	☐ Industrial Contamination	
☐ Operational Maintenance Check Issue	☐ Rain - Heavy	☐ Snow Pellets Showers - Heavy	☐ Insect Contamination	
☐ Physical Issue With Filter	☐ Rain Showers - Unspecified Intensity	☐ Snow Drains - Unspecified Intensity	☐ Internal Laboratory/Instrument Contamination	
☐ Power Failure	☐ Rain Showers - Light	☐ Snow Drains - Light	☐ Pollen/Leaf Contamination	
☐ Sample Diluted for Analysis	☐ Rain Showers - Moderate	☐ Snow Drains - Moderate	☐ Rain Showers - Unspecified	
☐ Unmeasured Key Meteorological Parameter	☐ Rain Showers - Heavy	☐ Snow Drains - Heavy	☐ Exceptional Event - Unspecified	
☐ Sample Not Analyzed	☐ Freezing Rain - Unspecified Intensity	☐ Diamond Out - Unspecified Intensity	☐ Seismic Activity	
☐ Operational Maintenance - Unspecified	☐ Freezing Rain - Light	☐ Diamond Out - Light	☐ Stratospheric Cruise Situation	

Statistics



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Statistical modes

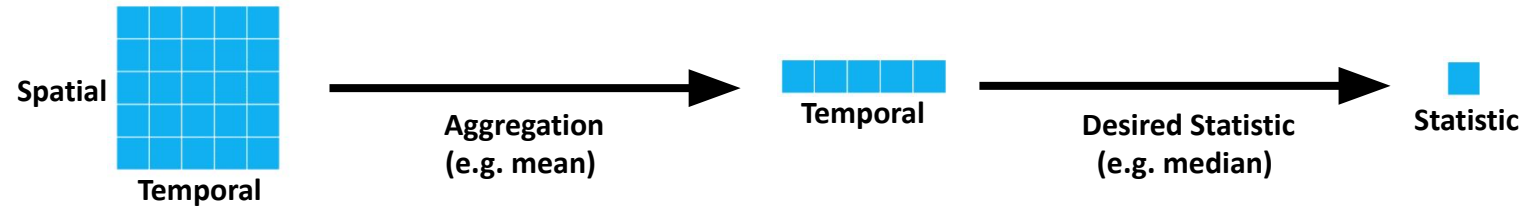
Most of the plotted data in Providentia depends on certain logical choices made when calculating statistics.

Statistics can be calculated in a variety of ways, and Providentia allows a number of ways to alter the calculation, via 3 statistical modes. These are as follows:

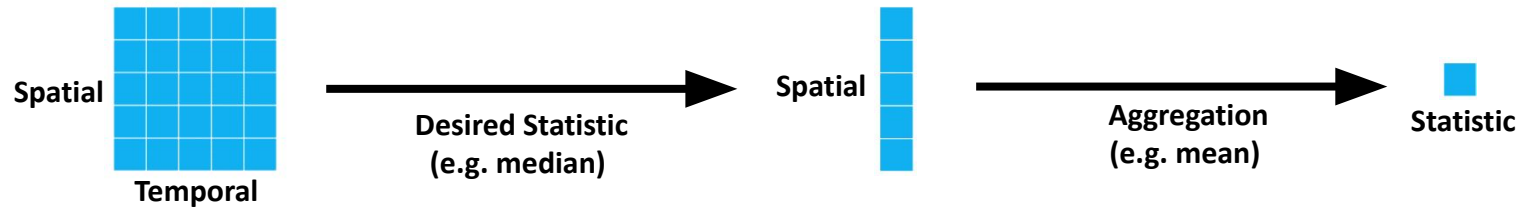
- **Temporal|Spatial (default)**
- **Spatial|Temporal**
- **Flattened**

Statistical modes

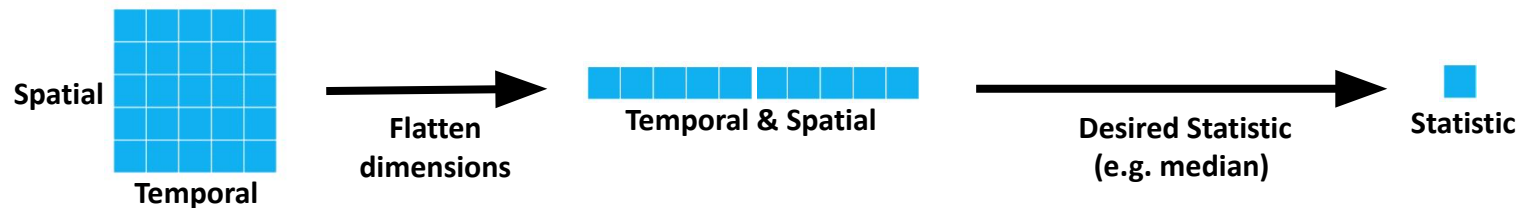
Spatial | Temporal



Temporal | Spatial



Flattened



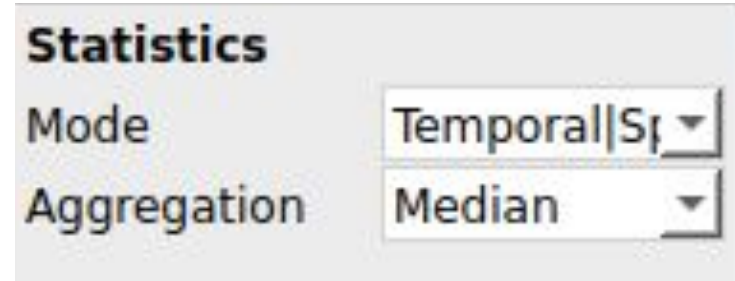
2D

1D

0D

Statistical modes

On the dashboard, in the statistics menu at the top, the mode and aggregation can be selected via the dropdown menus.



In the configuration files these can be set like so:

statistic_mode = Temporal|Spatial

statistic_aggregation = Median

Note: For the Flattened mode, there is no aggregation statistic.

BREAK
5 min



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Library



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Library

The library functionality allows Providentia to be used as a Python module, **with access to its backend functions**, allowing different parts of Providentia to be used in scripts, as required. In that way, we can import it:

```
import providentia as prv
```

From a Python script or notebook outside Providentia it is important to add the path to your directory to the system paths before importing Providentia:

```
import sys  
sys.path.append("path/to/your/Providentia/directory")
```

Examples: <https://github.com/BSC-ES/providentia/tree/master/tutorials>

Opening Jupyter notebook session



One typical use case is the interactive evaluation in a Jupyter notebook. Providentia has an in-built manner of opening a Jupyter notebook. If we add `--notebook` or `--nb` as a launch option in the command line, a Jupyter session will start:

```
./bin/providentia --notebook
```

Running locally, Jupyter **will open automatically**.

Loading a configuration file

Once imported the first step is calling the class Providentia with the configuration you plan to use.

```
provi = prv.Providentia("interactive_template.conf")
```

This class' methods can be used:

- To access the modes: download, interpolate, report and dashboard.
- To load, filter, save and plot the data.

You can specify the section or subsection you want to work with by. Otherwise, if you have multiple sections, the first one will be selected unless it is explicitly defined.

```
provi = prv.Providentia("interactive_template.conf",  
section="SECTIONNAME", subsection="SECTIONNAME.SUBSECTIONNAME")
```

Viewing active configuration

It is possible to view the active configuration variables in two different ways:

- Using the method *print_config*:

```
provi.print_config()
```

This method can also be used to view another configuration file, by passing the filename:

```
provi.print_config("important.conf")
```

- By printing the active class instance:

```
print(provi)
```

Accessing the modes

The *download*, *interpolate*, *report* and *dashboard* functionalities can be integrated in workflows with little effort.

```
provi.download()
```

```
provi.interpolate()
```

```
provi.report()
```

```
provi.dashboard()
```

Loading the data

When the data is available in your machine, you can load it by doing:

```
provi.load()
```

Once the data has been loaded, it can be easily accessed by these methods:

- Retrieving a specific variable, by providing a variable name to the *variable* method.

```
var_data = provi.variable("network|component_variablename")
```

- Retrieving all metadata / data variables by using the *data* method. The returned data can be in different formats: nc (netCDF), np (numpy) or xr (xarray) by passing the format argument:

```
data = provi.data(format="xr")
```

Applying filters

To apply a filter not set in the configuration file, the *filter* method is used. The fields to filter by can be any data /metadata variable standardised in GHOST, and can be textual or numeric.

- If the field is numeric, use *lower* and/or *upper*, e.g.:

```
provi.filter(field, lower=28, upper=31)
```

- If the field is textual, use *keep* and/or *remove*, e.g.:

```
provi.filter("country", keep="Spain")
```

```
provi.filter("country", remove=["Spain", "France"])
```

- If the field is a representativity field, use *limit*:

```
provi.filter(rep_field, limit=20)
```


Selecting stations

To select station for one or more stations, you can use the function `filter_station`:

```
provi.filter_station(['Kuwait_University_2'])
```

Reset filters

If at any time any active data / metadata filters are wished to be reset, this can be done by using the *reset* method.

To reset all filters to the Providentia defaults, including those set from the original read configuration file:

```
provi.reset()
```

To keep the filters defined in the configuration file, but reset the ones added later, pass the *initialise* argument as True:

```
provi.reset(initialise=True)
```

Plotting

You can use the plotting functions of Providentia with the *plot* method. Any of the available plot types in Providentia can be made by passing the desired type.

```
provi.plot(plot_type)
```

Plots can also be saved to file, by passing *save* and a filename with the wanted extension (e.g. png, etc.):

```
provi.plot(plot_type, save="myplot.png")
```

The plotting object can also be returned for additional tinkering, by setting *return_plot* to True:

```
plot_obj = provi.plot(plot_type, return_plot=True)
```

Reference: <https://providentia.readthedocs.io/en/latest/Plotting.html>

Saving the data

Save the current configuration:

```
provi.save(format='conf')
```

Save the data as netCDF:

```
provi.save(format='nc')
```

Save the data as numpy:

```
provi.save(format='np')
```

If you want to save the data to a specific location, use the argument `fname`. By default, the data is saved in the folder **providentia/saved_data**:

```
provi.save(fname='data/output/config.conf', format='conf')
```

Plot types



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Plot types

- Map (**map**)
- Metadata summary (**metadata**)
- Timeseries (**timeseries**)
- Periodic plot (**periodic**)
- Periodic violin plot (**periodic-violin**)
- Box plot (**boxplot**)
- Distribution plot (**distribution**)
- Scatter plot (**scatter**)
- Heat map (**heatmap**) - Not available in dashboard
- Table that gives one statistic per subsection per model (**table**) - Not available in dashboard
- Table that gives multiple statistics per model (**statsummary**)
- Taylor Diagram (**taylor**)
- FAIRMODE target plot (**fairmode-target**)
- FAIRMODE statistics summary plot (**fairmode-statsummary**)
- Contingency table (**contingencytable**)

Reference: <https://providentia.readthedocs.io/en/latest/Plot-types-and-options.html>

Statistics in plots

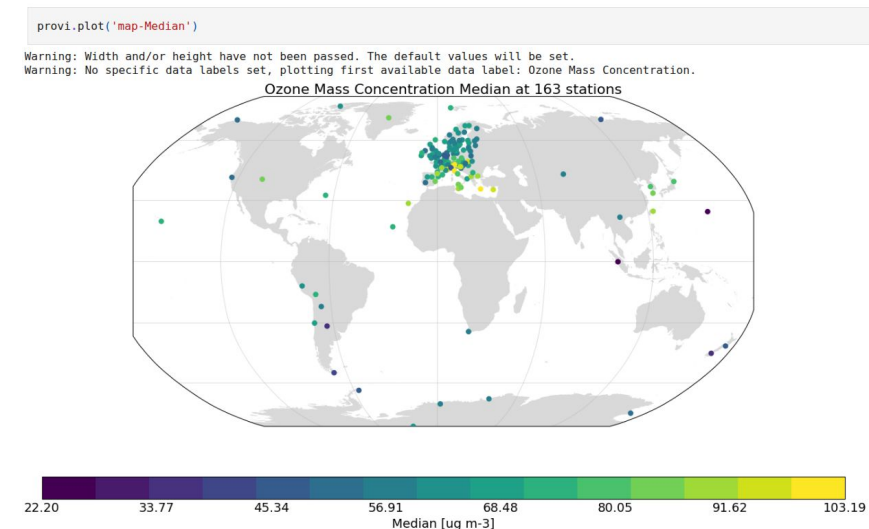
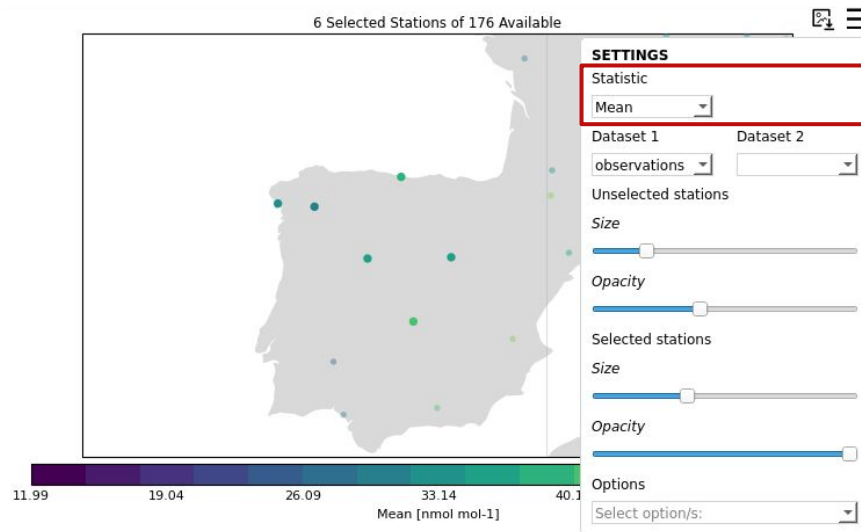


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Statistics in plots

It must be added as `-[stat]` (report, library) or selected from the burger menus (dashboard) to create **maps, periodic plots, heatmaps, tables and Taylor diagrams**. For example, if you want to see the mean / median of the observations in the dashboard (left) and library (right), we should do the following:



Reference:

<https://providentia.readthedocs.io/en/latest/Statistics.html#available-statistical-metrics>

Available statistics

Basic statistics

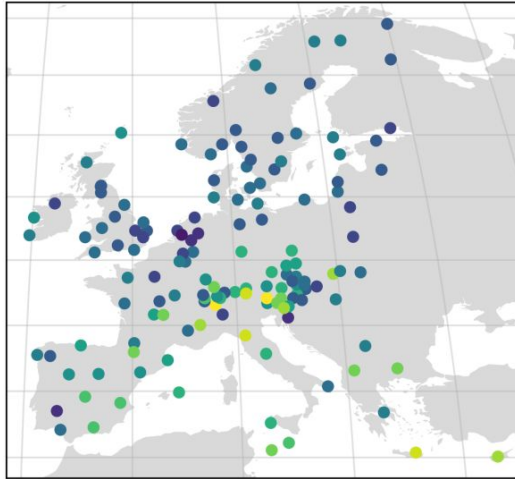
Statistic	Meaning
Mean	Mean
StdDev	Standard deviation
Var	Variance
Min	Minimum
Max	Maximum
Data%	Data availability
Exceedances	Number of exceedances
p1, p5, p10, p25, p50, p75, p90, p95, p99	Percentiles
NStations	Number of stations
MDA8	Daily maximum 8 hour average

Model bias statistics

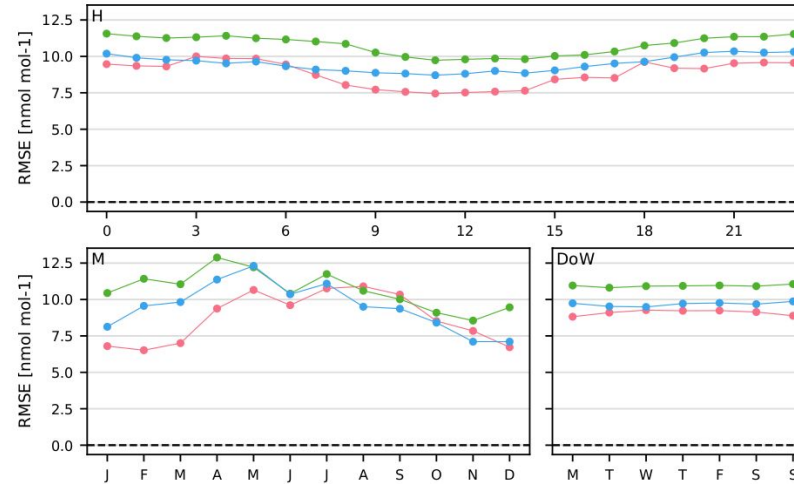
Statistic	Meaning
MB	Mean bias
NMB	Normalised mean bias
ME	Mean error
NME	Normalised mean error
MNB	Mean normalised bias
MNE	Mean normalised error
MFB	Mean fractional bias
MFE	Mean fractional error
RMSE	Root mean square error
NRMSE	Normalised root mean square error
COE	Coefficient of efficiency
FAC2	Fraction of model values within a factor of two of observed values
IOA	Index of agreement
R	Pearson correlation coefficient
R ²	Coefficient of determination
UPA	Unpaired peak accuracy

[plot type]-[stat] examples

map-Mean



periodic-RMSE



taylor-r

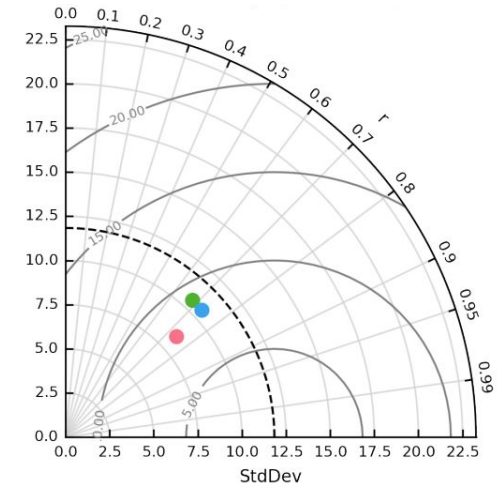
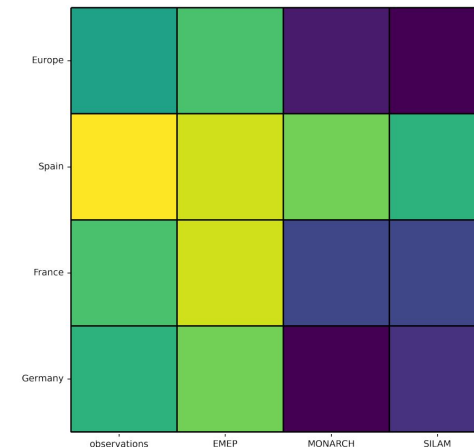


table-Data%

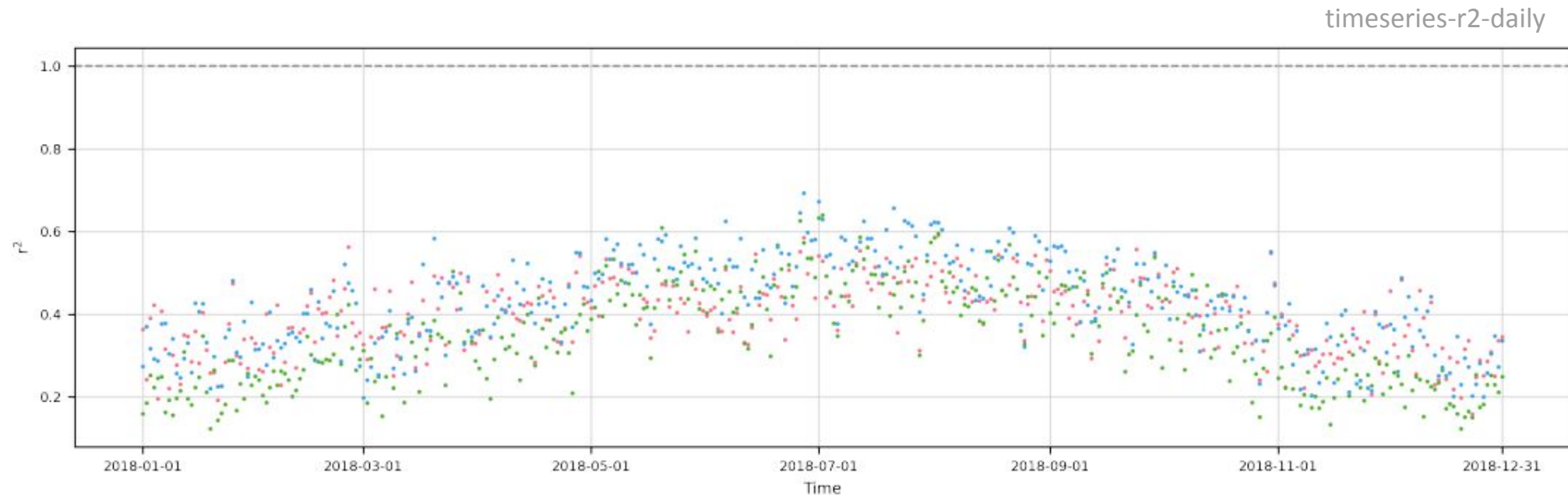
	observations	EMEP	MONARCH	SILAM
Europe	19.16	19.16	19.16	19.16
Spain	19.56	19.56	19.56	19.56
France	19.29	19.29	19.29	19.29
Germany	17.41	17.41	17.41	17.41

heatmap-Mean



timeseries-[stat]-[resolution]

The timeseries can also be used to show how statistics vary in time. In order to do this, we need to add **-[stat]** and the temporal resolution after the plot type name (e.g. *timeseries-Mean-daily*, *timeseries-r2-monthly*, *timeseries-r-annual*).



Customisation



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Plot customisation

To update the style of plots in the dashboard, reports and library, edit the plot characteristics in **settings/plot_characteristics.yaml**.

settings/plot_characteristics.yaml

All

Dashboard

Report

Library

```
"timeseries":
{
  "grid": { "axis": "both", "color": "lightgrey", "alpha": 0.8 },
  "threshold_line":
    { "linestyle": "--", "linewidth": 1.0, "color": "red", "zorder": 0 },
  ...
  "dashboard":
    {"remove_spines": ["top", "right"], ...},
  "report":
    {"orientation": "landscape", ...},
  "library":
    {"figsize": [17, 6], ...},}
```

Reference: <https://providentia.readthedocs.io/en/latest/Plot-customisation.html>



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Thank you for your attention!

More information at:

<https://github.com/bsc-es/providentia>

Join the #providentia Slack channel!

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